



The moderating effect of big data analytics on the international experience - post-entry internationalization speed relationship: An empirical study of U.S. public firms

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ABSTRACT

The accelerating digital transformation of international business has given rise to a critical tension: while traditional organizational learning theory emphasizes the gradual accumulation of experiential knowledge as the foundation for international expansion, Big Data Analytics introduces an alternative, algorithmic logic of learning that challenges this established paradigm. Grounded in organizational learning theory, this study examines how Big Data Analytics reconfigures the relationship between international experience, in both depth and breadth, and post-entry internationalization speed. We analyze panel data from 242 publicly listed US firms spanning 2010–2023, using a news-based measure of Big Data Analytics implementation. Our focus on large, established, non-IT public firms provides nuanced insights while bounding generalizability to this strategic context. For depth, Big Data Analytics mitigates the rigidity of deep experience, enabling firms to sustain faster PIS where returns would normally diminish, yet it simultaneously undermines this acceleration in early stages by suppressing the experiential learning vital for developing routines. For breadth, Big Data Analytics aids early cross-market learning but paradoxically undermines the strategic advantage of a broad portfolio, limiting the premium on extensive experience. Our study provides actionable insights for aligning Big Data Analytics with firms' experiential profiles to optimize post-entry internationalization trajectories.

1. Introduction

In today's global market, the speed of international expansion has emerged as a decisive factor in sustaining a competitive edge (Acedo & Jones, 2007). While prior research has extensively examined rapid internationalizers, such as Born Globals and International New Ventures (INVs) (i.e., Oviatt & McDougall, 2005; Weerawardena et al., 2007), growing scholarly attention has turned towards more established firms in later-stage internationalization. These firms often confront distinct challenges, such as competency traps (where routines become maladaptive in new contexts) (Leonard-Barton, 1992; Levitt & March, 1988) or the mismatch problem (where existing capabilities fail to align with idiosyncratic foreign market demands) (Fletcher et al., 2021; Perkins, 2014). For these established players, post-entry internationalization speed, defined as the rate of expansion after initial foreign market entry (Casillas & Acedo, 2013; Vermeulen & Barkema, 2002), has become increasingly critical (Hitt et al., 2016).

To explain how firms achieve such speed, research has long drawn on

organizational learning theory and its seminal application in the Uppsala model of internationalization. This paradigm rests on the premise that internationalization is fundamentally driven by experiential learning, where firms accumulate knowledge gradually through incremental market commitments to mitigate the liability of foreignness and strengthen relational networks (Johanson & Vahlne, 1977, 2009; Zaheer, 1995). This experience emerges from human engagement with foreign markets, generating situated insights that are subsequently codified into firm-level routines and institutionalized into organizational learning (Eriksson et al., 1997; Feldman & Pentland, 2003; March, 1991; Zollo & Winter, 2002). This process reflects substantive rationality, where reasoning and strategy emerge from values, situational meaning, and human judgment rather than purely formal rules (Dewey, 1922; Lindebaum et al., 2020). Consequently, international experience, particularly its depth (the intensity of experience within markets) and breadth (diversity of country exposure), has been viewed as the cornerstone of adaptation and expansion (Barkema & Drogendijk, 2007; Zeng et al., 2013).

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However, the rise of Big Data Analytics might challenge these established theoretical premises. Big Data Analytics refers to the end-to-end process of collecting, integrating, and analysing large-scale datasets using machine learning and other analytical techniques to navigate global market shifts (Chatterjee et al., 2023; Cheng et al., 2020; Gunasekaran et al., 2017; Xia et al., 2023). It embodies a competing logic of formal rationality (Balasubramanian et al., 2022), where learning is not gradual or socially embedded, but instead driven by data-intensive computation and rule-based inference (Broussard, 2018; Choudhury et al., 2020). By autonomously deriving decision patterns from correlations in large datasets, Big Data Analytics potentially marginalizes the contextual richness and routine diversity essential for adaptive, human-led learning (Choudhury et al., 2020).

Thus, the implementation of Big Data Analytics represents more than a technological enhancement; it introduces a fundamentally different logic of organizational learning, making the distinction between international experience and Big Data Analytics consequential. This is particularly salient in internationalization, where effective decision-making requires more than pattern detection; it demands sensitivity to local nuance, institutional heterogeneity, and learning from failure - capabilities traditionally rooted in human experiential knowledge. This algorithmic-experiential learning tension echoes concerns regarding the virtuality trap, where heavy reliance on digitally mediated information creates an illusion of knowledge that obscures tacit and relational cues essential for market understanding (Yamin & Sinkovics, 2006). Related work on cognitive digitalization biases similarly shows that firms often overestimate digital tools' ability to substitute for embodied, context-rich knowledge, leading them to underinvest in mechanisms for interpreting local conditions (Bouncken et al., 2023). Together, these insights suggest that algorithmic reasoning can distort, rather than simply enhance, experiential learning, highlighting the need to examine how Big Data Analytics interacts with international experience to shape post-entry internationalization speed.

Building on these concerns, the tension between algorithmic and experiential learning might challenge the core assumptions in organizational learning theory for experience-based learning in the International Business (IB) landscape in three ways. First, experiential learning is assumed to be path-dependent and locally embedded, emerging from situated interactions that crystallize into routines (Feldman & Pentland, 2003; Zollo & Winter, 2002). Big Data Analytics can bypass these lived processes, substituting them with decontextualized algorithmic inference. Second, experiential learning assumes selective retention based on human feedback (March, 1991), whereas Big Data Analytics shifts routine refinement toward automated statistical optimization, potentially narrowing routine diversity and undervaluing tacit and relational cues. Third, organizational learning theory highlights errors and local adaptation as catalysts for learning (Levitt & March, 1988), yet Big Data Analytics prioritizes predictive stability and anomaly suppression, filtering out the very irregularities from which experiential learning arises.

Therefore, these theoretical tensions lead to our central research question: *How does Big Data Analytics implementation reconfigure the effects of depth and breadth of international experience on post-entry internationalization speed?* Addressing this question is crucial for several reasons. First, prior moderators of the international experience - speed relationship, such as home regional concentration (Mohr & Batsakis, 2014) or domestic experience (Li et al., 2021), operate within traditional experiential logics. By contrast, Big Data Analytics introduces an external, algorithmic capability that might contest the Uppsala model's core assumptions of gradual, experience-based learning (Autio et al., 2021), and generate a contextualization gap by undervaluing tacit, context-specific knowledge. This also aligns with Argote et al.'s (2021) call to investigate how Big Data reshapes organizational learning by shifting the balance between intuition and data-driven reasoning. Second, the interaction between Big Data Analytics and international experience is unlikely to be uniform, as experiential breadth and depth

represent distinct learning pathways with different underlying mechanisms (Autio et al., 2000; Casillas & Moreno-Menéndez, 2014; J. W. Lu & Beamish, 2004). Examining these pathways reveals how firms reconcile human and algorithmic learning as they expand abroad, especially as firms increasingly integrate data-driven decision tools into their global strategies (Argote et al., 2021; Balasubramanian et al., 2022).

Using panel data from publicly listed US firms from 2010 to 2023, we theorize and empirically test how Big Data Analytics moderates the international experience - post-entry internationalization speed relationship through distinct pathways in depth-based and breadth-based experiential contexts. We argue that Big Data Analytics does not simply strengthen or weaken experiential effects but actively restructures the logic of experiential learning through two mechanisms identified by Balasubramanian et al. (2022): the selection effect, where algorithmic optimization narrows routine diversity, and the nescience effect, where reliance on codified patterns erodes tacit knowledge embedded in deep international engagement.

Our study makes several key contributions. First, it extends organizational learning theory by theorizing how Big Data Analytics reconfigures the Uppsala model's premise of incremental, experience-based learning (Johanson & Vahlne, 1977, 2009). In doing so, we specify the mechanisms (selection and nescience effects) by which Big Data Analytics' formal rationality can both complement and disrupt the gradual institutionalization of experience into routines (Feldman & Pentland, 2003; Zollo & Winter, 2002), addressing urgent calls to examine how digital technologies disrupt established theoretical frameworks (Autio et al., 2021; Verbeke & Hutzschenreuter, 2021). Second, we advance the Big Data Analytics scholarship by moving beyond performance-centric perspectives to theorize Big Data Analytics as a critical moderator that reshapes how firms utilize experiential knowledge in international contexts. This responds to Argote et al.'s (2021) calls for the integration of data-driven insights with human inputs in organizational learning. Third, we refine the understanding of post-entry internationalization speed by demonstrating how Big Data Analytics differentially moderates how experiential learning shapes expansion outcomes, positioning post-entry speed as jointly determined by experiential and data-driven learning processes.

2. Theoretical framework

Internationalization is a complex learning process shaped by the depth and breadth of international experience. With the rise of Big Data Analytics, such learning process is being potentially reshaped by the interaction of human learning and machine-driven insights. This section builds the theoretical foundation for our study by adapting the framework proposed by Balasubramanian et al. (2022) to explore how Big Data Analytics moderates the international experience - post-entry internationalization speed relationship.

2.1. The duality of learning from international experience

Internationalization hinges on organizational learning, where managers and employees acquire knowledge through interactions with foreign markets and internal artifacts, such as past entry records or operational data (Baum et al., 2000; Ingram & Baum, 1997; Johanson & Vahlne, 1977, 2009). This experiential learning process, rooted in human cognition and substantive rationality, requires individuals to interpret information through values, judgments, and contextual understanding (Dewey, 1922; Lindebaum et al., 2020). Such learning evolves into organizational learning through interactions among firm members, becoming embedded in collective routines as tacit knowledge, heuristics, or formalized procedures, shaping how firms expand after entering a foreign market (Argote & Miron-Spektor, 2011; Casillas & Moreno-Menéndez, 2014; Johanson & Vahlne, 1977). Experiential learning typically develops through two main mechanisms: repeated international activities and exposure to diverse markets, i.e., the *depth*

and *breadth* of international experience (Casillas & Moreno-Menéndez, 2014). Each mechanism has distinct implications for post-entry speed.

Depth of international experience enhances firms' familiarity with local dynamics, institutions, and stakeholders, reducing uncertainty and facilitating faster post-entry expansion (Eriksson et al., 1997; Johanson & Vahlne, 1977; Zaheer, 1995). Repeated engagement helps firms refine market-specific routines and build relational capital that accelerates subsequent commitments (Casillas & Moreno-Menéndez, 2014). However, deep experience may result in competency traps (Hannan & Freeman, 1984; Levitt & March, 1988), where familiar practices may stifle innovation and adaptability (Leonard-Barton, 1992; Schilke, 2014). This overreliance on familiar patterns may generate inferential errors, obscure emerging opportunities, and ultimately slow internationalization.

Breadth of experience exposes firms to diverse institutional and cultural environments, broadening their knowledge base and enhancing their ability to recognize patterns across markets (Casillas & Moreno-Menéndez, 2014; Eriksson et al., 1997, 2000). Yet, diversity also increases complexity. Early in the internationalization process, firms often struggle with cognitive overload and the mismatch problem, as insights from one context may not generalize to another (Barkema & Drogendijk, 2007; Hutzschenreuter & Voll, 2008; Schilke, 2014). Consequently, initial steps in a new market are often cautious and incremental, with slower subsequent operations (Andersen, 1993; Johanson & Vahlne, 1977). As breadth accumulates, however, firms develop sophisticated meta-routines for managing heterogeneity (Zhang et al., 2010). These higher-order learning capabilities improve their ability to transfer knowledge across contexts, enabling more efficient decision-making and accelerating subsequent international expansion (Casillas & Moreno-Menéndez, 2014; Eriksson et al., 1997).

2.2. Big Data Analytics as a new learning logic: affordances and limitations

Big Data Analytics involves computational methods used to process and analyse large datasets to derive actionable insights (Gandomi & Haider, 2015). Unlike human learning, which relies on interpretation and contextual judgment, it improves predictive accuracy by computing statistically "more correct" solutions, with "correctness" measured by quantitative metrics such as accuracy or model fit rather than contextual relevance (Balasubramanian et al., 2022; Lindebaum et al., 2020). This reflects a logic of formal rationality: statistical models trained on codified input data infer optimal decision rules that best predict outcomes (Broussard, 2018). Importantly, Big Data Analytics does not rely on a predefined, human-designed ontology; instead, it autonomously derives patterns from data, offering flexibility but also increasing opacity, particularly in complex model architectures such as neural networks (Negnevitsky, 2005; Provost & Fawcett, 2013). In practice, firms implement Big Data Analytics using prepackaged algorithms trained on historical data and validated before deployment, often with minimal human intervention (Broussard, 2018; Cui et al., 2006; Singh et al., 2025).

This unique approach to learning allows Big Data Analytics to play a significant role in shaping organizational learning processes by enabling the synthesis of disparate data streams, identification of structural patterns, and strategic choices informed by quantitative insights. Organizational learning theory emphasizes the importance of data integration and transfer to generate knowledge that improve firm decision-making through the acquisition, interpretation, and generation of knowledge (Levitt & March, 1988). By integrating huge volumes of data from diverse sources, Big Data Analytics supports these mechanisms by facilitating pattern recognition, evidence-based evaluation of alternatives, and the incorporation of external knowledge sources allows firms to uncover new patterns and solve problems effectively (Choo, 1996; Sivarajah et al., 2017; Xu et al., 2016).

Yet, while Big Data Analytics aligns with the learning mechanisms

emphasized in organizational learning theory, its effectiveness is tempered by two critical limitations proposed by Balasubramanian et al. (2022) when exploring the interplay between human learning and machine learning: the selection effect and the nescience effect. The selection effect arises because Big Data optimizes toward statistically superior routines, privileging one-best-way solutions over context-specific adaptations and potentially reducing the diversity of approaches for coping with heterogeneous environments (Balasubramanian et al., 2022; Pentland, 2005). A well-known example is Tesco's expansion in China: its reliance on analytics-driven standardization of the Clubcard program and product assortment, mirroring UK consumer patterns, failed to account for Chinese consumers' variety-seeking behavior, preference for bargaining, and smaller shopping baskets. This has ultimately contributed to Tesco's exit in 2020 (Park, 2016). The nescience effect, meanwhile, reflects Big Data Analytics' inability to capture uncodifiable, tacit, or relational knowledge (Balasubramanian et al., 2022; Choudhury et al., 2020). Elements such as cultural nuances, or relationship-based systems like Chinese *guanxi*, which are vital for successful internationalization (Chu et al., 2020; Du & Zhou, 2019), resist algorithmic representation. These informal institutional factors, which Peng (2003) identifies as central for navigating emerging market voids, are often better understood through experiential judgment than through computation.

3. Conceptual framework and hypotheses

Prior research establishes that international experience influences post-entry internationalization speed in non-linear ways: depth follows an inverted U-shaped pattern, while breadth exhibits a U-shaped pattern (Casillas & Moreno-Menéndez, 2014). In our study, we adopt these relationships as a baseline model to establish the foundational effects of international experience on post-entry internationalization speed. However, the central focus of our analysis lies in extending this model by examining how Big Data Analytics implementation moderates these effects and reshapes the way firms translate experience into post-entry internationalization speed. Before theorizing these moderating mechanisms, we first clarify the baseline roles of depth and breadth to ground our subsequent arguments.

3.1. International experience (depth and breadth) and post-entry internationalization speed

As firms deepen their commitment within specific foreign markets, they build familiarity with local institutions, customers, and networks, reducing the liability of foreignness (Eriksson et al., 1997; Johanson & Vahlne, 2009; Zaheer, 1995). This accumulated depth of experience through trial-and-error learning allows firms to navigate host-country environments more efficiently, accelerating post-entry internationalization speed (Gao & Pan, 2010). However, firms encounter diminishing returns from new foreign entries with increasing levels of international experience, as additional internationalization offers little novel insight (Thakur-Wernz & Samant, 2019). Once experiential knowledge becomes overly internalized, it may generate competency traps (Levitt & March, 1988), where familiar routines hinder adaptability and slow further expansion. Thus, we expect an inverted U-shaped relationship between international experience depth and post-entry internationalization speed.

In contrast, increasing the breadth of international experience by entering many diverse markets initially imposes significant cognitive and coordination challenges, slowing down post-entry internationalization speed. This occurs because each new market entry introduces a fresh liability of foreignness (Zaheer, 1995), requiring firms to navigate unfamiliar institutional environments from the beginning. This also creates increasing cognitive and operational burdens, requiring substantial resources to acquire and absorb knowledge (Eriksson et al., 1997; Vermeulen & Barkema, 2002). At such breadth levels, firms face

extensive heterogeneity that prevents generalization, yet insufficient accumulated diversity to develop transferable routines across markets (Zaheer, 1995). Experimentation across different contexts delays the formation of standardized processes and routines (Nadkarni & Perez, 2007), producing a negative relationship between lower levels of breadth and post-entry speed. However, as firms reach moderate-to-high breadth, they develop stronger capabilities to identify common patterns across markets and build meta-routines adaptable to multiple contexts (Barkema et al., 1996). Furthermore, firms with broader prior experience have built strong absorptive capacity (Cohen & Levinthal, 1990), improving knowledge transfer and reducing the time and resources required for each new market entry. Therefore, the relationship between breadth and post-entry internationalization speed is expected to follow a U-shaped pattern. Taken together, we have the baseline models as below:

H0a. There is an inverted U-shaped relationship between depth of international experience and post-entry internationalization speed.

H0b. There is a U-shaped relationship between breadth of international experience and post-entry internationalization speed.

3.2. The joint effect of Big Data Analytics implementation and depth of international experience on post-entry internationalization speed

The interaction between deep, experience-based learning and algorithmic Big Data Analytics is governed by a fundamental clash between the organizational learning theory's premise of knowledge rooted in situated, path-dependent interpretation (substantive rationality) and Big Data Analytics' logic of algorithmic optimization (formal rationality). This tension materializes through the selection and nescience effects (Balasubramanian et al., 2022). This mismatch determines whether Big Data Analytics implementation results in a net effect of synergy (mitigating experiential rigidity) or displacement (suppressing essential trial-and-error learning). We structure our argument by showing how the selection and nescience effects operate within this central tension at different stages of experiential depth.

3.2.1. Low-to-moderate depth of international experience stage

At low-to-moderate levels of experiential depth, the firm's learning is governed by the logic of substantive rationality: it is actively engaged in trial-and-error, building context-sensitive routines from repeated interactions with the market environment (Dewey, 1922; Eriksson et al., 1997; Pentland, 2005). The integration of high-level Big Data Analytics at this stage introduces a powerful competing logic of formal rationality. This tension triggers the selection and nescience effects, with Big Data Analytics dominating over early experiential depth for two key reasons. First, Big Data Analytics' real-time, large-scale data which offers immediate and statistically precise insights often takes precedence over the slower, fragmented knowledge derived from nascent international experience (Cui et al., 2006; Mikalef et al., 2019). Second, high levels of Big Data Analytics implementation involves the institutionalization of data-driven logic through formalized processes, such as KPI dashboards and automated reporting, that shape organizational attention to codified outputs (Feldman & Pentland, 2003; Orlikowski & Barley, 2001). As a result, algorithmic prescriptions are likely to be privileged over contextual judgment.

Critically, this dynamic is amplified by what Bouncken et al. (2023) identify as cognitive digitalization biases. Firms tend to overestimate the ease and usability of digital technologies while underestimating the deep-level challenges rooted in the contextuality of international operations. The selection effect operates potently in this context. Big Data Analytics' algorithms, optimized to identify statistically dominant patterns from available data, narrow the range of strategic options deemed viable (Balasubramanian et al., 2022). While this reduces uncertainty, it also truncates the very trial-and-error learning process essential for building nuanced understandings of foreign markets. Managers,

confronted with decontextualized optimal choices, may bypass their emerging experiential insights (Balasubramanian et al., 2022; Feldman & Pentland, 2003), thereby diminishing the value of such experiential insights in shaping post-entry internationalization speed. In extreme cases, heavy reliance on Big Data Analytics can lock firms into impersonal rigid data-driven routines (Gligor et al., 2021), further stifling the adaptive learning critical for developing contextual knowledge. This selective narrowing reduces the marginal gains from each incremental unit of depth, thereby flattening the positive slope of the inverted U-curve.

Simultaneously, the nescience effect becomes more pronounced, creating a modern incarnation of the virtuality trap (Yamin & Sinkovics, 2006). Big Data Analytics' reliance on codifiable data creates systematic blind spots to the tacit, uncodifiable insights (e.g., institutional nuances, unwritten norms) that managers begin to internalize through early depth experiences (Lindebaum et al., 2020; Mintzberg, 1987). As Bouncken et al. (2023) note, digital technology often fails to cope with the "stickiness" of such knowledge, which requires socio-emotional sharing and direct personal co-presence. The firm, impressed by Big Data Analytics' statistically precise output, may mistakenly assume sufficient market knowledge has been acquired, under-investing in other non-virtual, on-site sources of information (Yamin & Sinkovics, 2006). Thus, firms systematically underestimate the potential returns from further deepening experience, leading them to accumulate experiential depth more slowly. This delays the point at which depth-based learning generates maximal benefit, effectively shifting the perceived turning point of the inverted U-curve rightward.

Together, these dynamics attenuate the positive effect of low-to-moderate experiential depth on speed, as the learning process that should accelerate post-entry speed is disrupted before it matures.

3.2.2. Moderate-to-high depth of international experience stage

At moderate-to-high levels of international experience depth, the logic of substantive rationality risks becoming a liability: routines are institutionalized, cognitive frames rigidify, and diminishing returns emerge (Leonard-Barton, 1992; Levitt & March, 1988). In this setting, high-level Big Data Analytics implementation can function as a corrective force. The selection and nescience effects manifest not as disruptors but as mechanisms that counter experiential lock-in, thereby mitigating the diminishing returns of experiential depth.

Specifically, the selection effect counteracts experiential lock-in by supplying algorithmically generated alternatives that broaden strategic possibilities. Big Data Analytics can surface anomalies that contradict entrenched heuristics, reveal latent growth opportunities and expand option sets (Brynjolfsson & McElheran, 2016). This is evident when Starbucks used AI-powered analysis of Chinese consumer behavior (e.g., WeChat engagement, digital ordering patterns) and partnership with Alibaba ecosystems to override its longstanding heuristic of standardized premium cafés, prompting localized adaptations such as tea-infused beverages and digitally-optimized store formats (Starbucks, 2018; The Strategy Institute, 2024). This data-enabled shift partly facilitated its rapid expansion to more than 7000 locations across 250 Chinese cities (Business Insider, 2024), illustrating how algorithmic insights can disrupt deep but path-dependent experiential frames and restore adaptability. Furthermore, high levels of Big Data Analytics also fosters a pluralistic, data-driven sensemaking that integrates diverse data sources and stakeholder perspectives (George et al., 2016; Popović et al., 2012), enabling firms to reinterpret past experiences through a more granular lens and revealing overlooked elements (Strike & Rerup, 2016), which in turn can mitigate cognitive lock-in and enhance strategic flexibility. Together, this flattens the negative slope of the curve, making the decline in returns from excessive depth less severe.

Moreover, the nescience effect offers an external, de-biased perspective invisible to managers constrained by deep experiential frames (Tripsas & Gavetti, 2000). Since Big Data Analytics does not encode tacit assumptions embedded in deep experience, it avoids

reproducing existing cognitive biases. By offering disconfirming, data-driven insights that loosen rigid experiential frames (Lindebaum et al., 2020), Big Data Analytics encourages firms to challenge institutionalized routines and stimulate more adaptive sensemaking.

Thus, at moderate-to-high depth, Big Data Analytics' formal rationality counterbalances the rigidity of substantive rationality, mitigating its declining returns on post-entry internationalization speed.

H1. The inverted U-shaped relationship between the depth of international experience and post-entry internationalization speed becomes less pronounced (i.e., the slope flattens) under high levels of Big Data Analytics implementation.

3.3. The joint effect of Big Data Analytics implementation and breadth of international experience on post-entry internationalization speed

The impact of experiential breadth rests on the firm's capacity for cross-contextual knowledge transfer while maintaining routine diversity. Big Data Analytics introduces a core theoretical tension that we term algorithmic overconvergence, where the drive for statistical efficiency (formal rationality) tends to homogenize the context-specific insights gained from diverse international exposure (substantive rationality). The selection and nescience effects illustrate the two aspects of this tension across stages of experiential breadth, either by efficiently generating transferable templates or by introducing destructive epistemic frictions that undermine the strategic value of experiential breadth.

3.3.1. Low-to-moderate breadth of international experience stage

At low-to-moderate levels of international experience breadth, firms operate under the logic of substantive rationality but with a fragmented, incomplete knowledge base across diverse markets (Barkema & Drogendijk, 2007; Hutzschenreuter et al., 2011), facing high cognitive and operational complexity with each new entry. High-level Big Data Analytics implementation at this stage introduces formal rationality that complements this fragile experiential base, where the selection and nescience effects function as stabilizing mechanisms, enhancing the efficiency of early breadth for post-entry speed.

Through the selection effect, Big Data Analytics narrows decision sets to statistically viable options based on cross-market data patterns. This helps reducing costly trial-and-error processes that typically characterize this stage of internationalization (Casillas & Moreno-Menéndez, 2014; Nadkarni & Perez, 2007) and enables the formation of generalized templates that support initial cross-border learning. Big Data Analytics can identify transferable practices across seemingly dissimilar contexts (Brynjolfsson & McElheran, 2016), helping firms overcome the "mismatch problem" and accelerating expansion speed. Supporting evidence from Sumbal et al. (2017) suggests that Big Data Analytics synthesizes multi-regional data to detect structural similarities, allowing firms to apply insights across regions while curtailing operational costs. Consequently, the initial disadvantages of low-to-moderate breadth are attenuated, flattening the negative slope of its U-shaped relationship with post-entry speed.

However, the nescience effect becomes more consequential for firms at this stage. Because Big Data Analytics relies primarily on codifiable correlations extracted from a still-limited cross-national base, it tends to overgeneralize shallow patterns and treat them as broadly applicable. Managers, whose mental models remain fragmented and prone to heuristic biases as they extrapolate from insufficiently diverse, often non-representative set of experiences (Tripsas & Gavetti, 2000; Tversky & Kahneman, 1974), may interpret these algorithmic outputs as confirmation of oversimplified schemas about foreign markets. As Bouncken et al. (2023) caution, digital technologies can channel interactions towards more generic "handshake" relationships rather than the deep "conversations" needed to understand contextual nuance. This can slow the accumulation of genuine context-sensitive insights, manifesting as a

deeper trough and a rightward shift in the turning point of the U-shaped relationship, as firms might require exposure to a larger number of countries before the benefits of breadth materialize in faster post-entry internationalization.

3.3.2. Moderate-to-high breadth of international experience stage

At moderate-to-high levels of international experiential breadth, firms possess a rich, diverse portfolio of market-specific insights - the pinnacle of substantive rationality in internationalization (Barkema et al., 1996; Bingham et al., 2007; Casillas & Moreno-Menéndez, 2014). However, high levels of Big Data Analytics implementation introduce a formal rationality that clashes with this nuanced, contextual knowledge base, creating epistemic friction, undermining the strategic value of broad international experience.

The selection effect becomes a source of over-convergence for firms at this stage. Big Data Analytics' optimization logic favours standardized, data-validated approaches that statistically dominate (Balasubramanian et al., 2022), inherently marginalizing the rare, context-specific, or tacit insights that constitute the value of a broad experiential portfolio. For instance, a niche strategy recognized by managers for long-term value might be deprioritized by Big Data Analytics due to weak short-term metrics. This algorithmic narrowing reduces routine diversity and weakens the advantages of broad experience, attenuating the positive effect of moderate-to-high breadth on post-entry internationalization speed. Crucially, by diluting the unique competitive advantages gained from a diverse global footprint, Big Data Analytics reduces the marginal speed gained from each additional unit of breadth, leading to a flattening of the positive slope of the U-shaped relationship as firms struggle to translate their broad knowledge into accelerated expansion speed.

Furthermore, the nescience effect exacerbates this by creating epistemic dissonance. When Big Data Analytics systems generate insights that contradict managers' nuanced, cross-contextual intuitions born of diverse exposure, it erodes confidence in both sources of insights and can cause decision paralysis (Wilson et al., 2018). Managers may become more risk-averse, defaulting to incremental adjustments or delaying critical decisions until the conflict is resolved. Instead of serving as a guide for action, high Big Data Analytics may dilute the authority of experiential insights, compelling firms to reconcile divergent cues under time pressure and uncertainty (Gavetti & Levinthal, 2000; Ocasio et al., 2021). Such contradictions can lead to internal debate or reliance on lowest-common-denominator solutions that fail to capitalize fully on either experience breadth or Big Data Analytics. Additionally, reconciling these logics demands costly integration mechanisms, such as ambidextrous learning structures or local model adaptation, which many firms lack (Fang et al., 2010).

In sum, at moderate-to-high breadth, the effects of Big Data Analytics flatten the positive slope and disrupt the location of the turning point, weakening the positive effect of broad experience on speed.

H2. The U-shaped relationship between breadth of international experience and post-entry internationalization speed becomes less pronounced (i.e., the slope flattens) under high levels of Big Data Analytics implementation.

4. Methods

4.1. Data and sample selection

Since our focal variable of interest - Big Data Analytics implementation - is measured through systematic analysis of corporate news coverage, we necessitate a sampling strategy that ensures sufficient and consistent strategic disclosure for all firms in our panel. Therefore, we draw our initial sample from the Fortune 500 firms. This focus is crucial to ensure the measurement rigor of our BDA variable. Our BDA measure aggregates firm-originated announcements (e.g., press releases,

company disclosures) regarding concrete BDA initiatives. Large, publicly scrutinized firms, like those in the Fortune 500, operate under significant institutional pressure and market incentives to issue public disclosures for major strategic investments. By sampling this group, we mitigate the critical risk of non-disclosure bias - the chance that a smaller, private, or less-scrutinized firm implemented BDA but simply did not announce it publicly. Furthermore, while our measure isolates firm-initiated news, the selection of the Fortune 500 provides an essential baseline of extensive and consistent media coverage (Joshi et al., 2010), ensuring that when an announcement is made, it is systematically captured in our search pool. To ensure theoretical coherence and data integrity, we apply several exclusion criteria. Firms in the financial and utility sectors are excluded due to their distinct regulatory environments and reporting requirements. We exclude IT firms, as their heavy reliance on IT could obscure the influence of Big Data Analytics implementation (Huang et al., 2020), and private firms due to the unavailability of their financial data. Firms classified as Born Globals or INVs (those that commenced internationalization within three years of establishment) and SMEs (those with < 250 employees) are excluded due to our research's focus (Gallego & Casillas, 2014; Knight & Cavusgil, 2004).

We collect secondary data from a variety of sources. We rely on secondary data from a wide range of news articles to examine firms' Big Data Analytics implementation, avoiding the biases associated with primary data, like surveys (Devaraj & Kohli, 2003). Although secondary data may be affected by media bias due to uneven coverage across industries, we mitigate this by sourcing information from over 4000 news outlets using LexisNexis and Factiva from 2010 to 2023. We choose 2010 as the starting year because it marked the rise of significant media attention on Big Data Analytics (Davenport, 2014; Huang et al., 2020), making pre-2010 and post-2010 data less comparable. Data on the depth and breadth of international experience and post-entry internationalization speed are collected from annual reports, and the Orbis database, excluding subsidiaries in tax havens (Lu et al., 2014). Patent data are sourced from the Intellectual Property Orbis database, while firm-level variables and industry characteristics are obtained from COMPUSTAT and 10-K filings. After integrating data using unique company codes and excluding unusable observations, we obtain an unbalanced panel dataset of 2854 firm-year observations from 242 firms.

4.2. Measures

4.2.1. Dependent variable

We measure *post-entry internationalization speed* as the number of foreign subsidiaries by the number of years since the firm's initial entry into foreign markets (Li et al., 2021; Vermeulen & Barkema, 2002). This metric captures both the extent and timing of foreign investments, with higher values reflecting faster internationalization.

4.2.2. Independent variables

Following Clarke et al. (2013), we conceptualized *depth of international experience* as encompassing both the length (duration) and intensity (commitment) dimensions, with our measure capturing cumulative overseas subsidiary ages (length) and implying deeper commitments in older subsidiaries (intensity). Following Casillas and Moreno-Menéndez (2014), we measured *breadth of international experience* by computing the coefficient between the sum of the squared frequency of operations in each country prior to the focal operation and the total number of operations before squaring. Given that a lower Herfindahl index value reflects greater diversity, we employ its inverse as the measure.

4.2.3. Moderating variable

Big Data Analytics implementation is measured by the cumulative count of annual Big Data Analytics announcements per firm (Huang et al., 2020; Joshi et al., 2010). Relevant news articles are identified by

searching for company names alongside Big Data Analytics-related keywords adopted from Huang et al. (2020) and international expansion terms (Ding et al., 2021). For each qualified article, the starting year of implementation is recorded; missing years are retrieved through additional keyword searches (e.g., analytics software names). Articles without recoverable starting years are excluded. Duplicates across outlets are removed through a rigorous cross-checking process, ensuring that only distinct instances of Big Data Analytics implementation are counted. Articles covering multiple firms are counted separately. Importantly, this measure captures firm-initiated news, including press releases and direct company disclosures, rather than media-initiated reports or third-party commentary. Following prior research (Dang et al., 2021; Huang et al., 2020; Tsileponis et al., 2020), such firm-originated announcements often serve as credible signals of actual strategic implementation. A manual content review confirms that nearly all qualified articles originate from firm communications and covered concrete initiatives (e.g., analytics systems, AI deployments, strategic partnerships). Thus, rather than reflecting ambient media attention or sentiment, this approach isolates and aggregates observable, intentional acts of Big Data Analytics implementation. An example is "Schlumberger announced an enterprise-wide deployment of the cloud-based DELFI cognitive EP environment for ConocoPhillips. ConocoPhillips will use Schlumberger digital solutions enabled by the DELFI environment to bring its reservoir engineering modeling, data and workflows to the cloud. 'This digital platform will drive workflow and data efficiency, enabling ConocoPhillips to meet their diverse reservoir engineering modeling needs worldwide,' said Rajeev Sonthalia, president, Digital Integration, Schlumberger. 'Expert teams from both companies will work closely together to integrate our cloud-based digital solutions and AI with ConocoPhillips' reservoir engineering modeling and workflows.'" A more detailed description of the search strategy can be found in the Appendix.

4.2.4. Control variables

Drawing on prior literature, we included a set of firm-level, industry-level and country-level control variables to mitigate potential confounding effects. At the firm level, we incorporate *firm size*, quantified as the logarithm of the number of employees (Darnall et al., 2010) since larger firms possess stronger capabilities and resources for internationalization (Li et al., 2021). As older firms tend to exhibit higher commitment to internationalization (Lyles et al., 2014), we control for *firm age*, measured as the logarithm of years since inception. We also include *Return on Assets (ROA)* from the previous year, as higher financial performance supports faster international growth (Anand et al., 2016). Another control variable is firms' *financial leverage*, calculated as the sum of cash equivalents, marketable securities, and accounts receivables divided by the firm's current liabilities (Batsakis & Theoharakis, 2021). Given that international expansion represents an entrepreneurial act (Ellis, 2011), we control for *entrepreneurial orientation*, measured from annual reports as the product of innovativeness, proactiveness, and risk-taking counts, divided by total words and normalized 0–1 (Hunt, 2021). We also control for *innovation*, measured as the logarithm of the total patents, as innovation-active firms are more likely to internationalize quickly (Hsieh et al., 2019). Moreover, we include three governance structure characteristics: *CEO duality* (1 if the CEO is also the chair and 0 otherwise), *board size* (number of directors), and *board independence* (percentage of independent directors) due to their influence on firm strategies (Sherman et al., 1998; Wang et al., 2022). At the industry-level, we include *competitive intensity* using the Herfindahl-Hirschman Index (HHI) (Li et al., 2008), operationalized as (1-HHI), with higher values indicating greater competition. At the country-level, *institutional distance* is calculated as the absolute value of the difference between the average score of a firm's foreign entries and the USA's score (Li et al., 2021), using the World Governance Indicators from the World Bank (Ang et al., 2015). We also control for *cultural distance*, measured with Hofstede's four dimensions and combined using Kogut and Singh's (1988) method. Finally, we include dummy variables

Table 1

A summary of the theoretical arguments for Hypotheses 1 and 2.

	Stage	Selection effect	Nescience effect	Net result
Depth of international experience	Low-to-moderate	Detrimental: Big Data Analytics prioritizes statistically optimal strategies, suppressing experiential trial-and-error learning needed to refine emergent routines and reducing the value of experiential learning.	Detrimental: Big Data Analytics overlooks tacit insights gained through early depth experiences, causing them to undervalue further deepening and accumulate experiential depth more slowly.	Weakening the positive relationship
	Moderate-to-high	Constructive: Big Data Analytics challenges rigid routines by introducing novel patterns and data-driven alternatives, reducing path dependence.	Constructive: Big Data Analytics prevents the replication of human cognitive biases that accumulate through deep experiential learning	Mitigating the negative relationship
Breadth of international experience	Low-to-moderate	Constructive: Big Data Analytics provides a data-driven roadmap for expansion, reducing the costly trial-and-error that characterizes this stage and enhancing cross-border knowledge transferability.	Constructive: Big Data Analytics' reliance on patterns promotes overgeneralization and premature routine lock-in, requiring broader country exposure before experiential breadth accelerates internationalization.	Mitigating the negative relationship
	Moderate-to-high	Detrimental: Big Data Analytics' optimization logic favours standardization over nuance, creating an algorithmic overconvergence that crowds out diverse, context-specific routines that are the true source of strategic advantage.	Detrimental: Big Data Analytics, by excluding tacit knowledge, creates epistemic friction and organizational dissonance, which leads to decision paralysis as algorithmic outputs clash with managers' nuanced, context-specific expertise.	Weakening the positive relationship

to account for unobserved industry- and time-specific effects. A summary of variables, measurements, and data sources is provided in Table 2.

4.3. Estimation method

Due to the path-dependence effect of internationalization speed (Casillas & Moreno-Menéndez, 2014; Johanson & Vahlne, 1977), we use dynamic models with a lagged dependent variable. To determine the appropriate lag structure, we compare the mean square error across models with first-, second-, and third-order lagged dependent variables. The results indicate that the first-order lag is sufficient, as its coefficient fully captures all past information. A primary econometric concern in this model is the potential endogeneity of our key explanatory variables: depth of international experience, breadth of international experience and Big Data Analytics implementation. These factors may be influenced by internationalization speed or shared unobserved shocks, such as fluctuating financial resources. Furthermore, reverse causality remains a risk even when using lagged predictors.

To address these concerns, we use system generalized method of moments (GMM) (Arellano & Bond, 1991; Arellano & Bover, 1995). This estimator is preferred over conventional panel ordinary least squares (OLS) or within-group estimation methods (Blundell et al., 2000; Blundell & Bond, 1998), both of which can yield inconsistent and biased estimates in dynamic panels. OLS generally neglects unobserved time-invariant firm effects, leading to an upward bias, while the within-groups estimator suffers from Nickell bias due to the correlation between the lagged dependent variable and the error term (Hsiao, 2022; Nickell, 1981). Consequently, the OLS and within-groups estimates for the lagged dependent variable serve as the upper and lower bounds, respectively; a valid GMM estimate should ideally fall between these two limits (Bond et al., 2001). System GMM offers several advantages over traditional panel data techniques: it accounts for unobserved firm-specific effects, includes a lagged dependent variable to capture the dynamic nature of internationalization speed, and, most importantly, uses internal instruments to control for the endogeneity of all explanatory variables.

Compared to the difference GMM estimator (Arellano & Bond, 1991), system GMM is more effective by assuming that the initial differences in the instruments are uncorrelated with fixed effects, thereby allowing for additional instruments in the level equation (Roodman, 2009). This is particularly advantageous when the data series is persistent or behaves similarly to a random walk - conditions under which difference GMM often exhibits significant finite sample bias (Huang

et al., 2020; Slesman et al., 2021). Moreover, if instruments are weak, the difference GMM estimator can result in a downward bias even more severe than that of the within-groups estimator (Blundell & Bond, 2000). System GMM simultaneously estimates regressions in both levels and differences, using lagged levels as instruments for the differenced equations and lagged differences as instruments for the level equations. Our final specification employs a two-step system GMM configuration to further enhance efficiency and address finite sample bias (Blundell & Bond, 2000). Finally, to mitigate multicollinearity when estimating interactions, all independent and moderating variables are mean-centered.

5. Results

5.1. Baseline results

Descriptive statistics and correlations for the study sample are presented in Table 3. As shown, most explanatory variables are not highly correlated. To assess multicollinearity, we calculate the Variance Inflation Factor (VIF) for all variables across models. All VIF scores fall well below the threshold of 10 (Hair et al., 2009), indicating that multicollinearity is not a concern.

Table 4 presents the two-step system GMM results. We confirm the validity of our GMM models using two diagnostic tests. First, the Arellano-Bond AR(2) test yields p-values > 0.05 for all models, indicating no second-order autocorrelation and confirming correct moment specification (Arellano & Bond, 1991; Roodman, 2009). Second, the Hansen J-statistics for all models produce p-values ranging between 0.1 and 0.9, indicating that we cannot reject instrument exogeneity. This validates our instrument set and the overall reliability of the GMM model (Hansen, 1982). In Table 4, Model 1 includes only control variables. In Models 2 through 5, we introduce first- and second-order depth and breadth of international experience. Models 6–8 examine the moderating role of Big Data Analytics.

5.1.1. Regression results

Hypothesis H0a predicts an inverted U-shaped relationship between depth of international experience and post-entry internationalization speed. Following the three-step procedure by Haans et al. (2016), we find robust support for this hypothesis. First, as shown in Model 3, the coefficient of the linear term of depth is significant and positive ($\beta = -0.017$, $p = 0.000$), while the squared term is negative and significant ($\beta = 0.003$, $p = 0.000$), consistent with an inverted U-shape. Second, the utest command (Lind & Mehlum, 2010) confirms that the slope at the

Table 2
Variables and data sources.

Variable	Description	Source
Post-entry internationalization speed	the number of foreign subsidiaries to the number of years since the firm's initial entry into foreign markets	Compustat and Securities and Exchange Commission (SEC) filing
Depth of international experience	the natural logarithm of the total ages of the parent firm's subsidiaries across all countries outside the host country	Compustat and SEC filing
Breadth of international experience	the inverse coefficient by summing the squares of operations in each country prior to the focal operation and dividing by the total number of operations before squaring	Orbis
Big Data Analytics implementation	the natural logarithm of the cumulative count of Big Data Analytics announcements made by each firm annually	Factiva and Lexis-Nexis
Financial leverage	the ratio of a firm's total debts to its total assets	Orbis
Firm age	the natural logarithm of years since inception, is controlled as older firms tend to have higher commitment in internationalization	Orbis
Firm size	the natural logarithm of the number of employees	Orbis
ROA	the percentage ratio of a firm's income to average total assets	Orbis
Board independence	the percentage of independent directors	Compustat
Board size	the total number of directors	Compustat
Institutional distance	the absolute value of the difference between the average score of a firm's foreign entries and the USA's score	World Governance Indicators
Cultural distance	the degree of cultural differences between the home and host countries based on Hofstede's four dimensions	Hofstede
Competitive intensity	the sum of the square of each firm's market share within the market	Compustat
Innovation	the natural logarithm of total patents a firm has	Orbis Intellectual Property
Entrepreneurial orientation	the product of innovativeness, proactiveness, and risk-taking counts, divided by the total number of words in annual reports, normalizing from 0 to 1	Annual reports from SEC filings
CEO duality	a binary variable which equals 1 if the CEO is also the chair, 0 otherwise	Compustat

low end (X_L) is positive and significant ($X_L = 0.040$, $p = 0.000$), and the slope at the high end (X_H) is negative and significant ($X_H = -0.047$, $p = 0.000$). The overall test for an inverted U-shape is highly significant ($p = 0.000$). Third, the calculated turning point of the curve (-2.05 on the centred scale) falls well within the observed data range, satisfying the final condition for a valid inverted U-shape.

Hypothesis H0b posits a U-shaped relationship between breadth of international experience and post-entry internationalization speed. Model 4 supports this prediction. The linear term for breadth is negative and significant ($\beta = -0.458$, $p = 0.000$), while the squared term is positive and significant ($\beta = 0.416$, $p = 0.021$). The utest results confirm the U-shape ($p = 0.000$), with a significantly negative slope at the lower bound ($X_L = -0.79$, $p = 0.000$), and a significantly positive slope at the upper bound ($X_H = 0.72$, $p = 0.000$). The turning point (0.33) falls within the observed data range. Model 5, which includes both quadratic terms simultaneously, confirms these non-linear patterns. The Sasabuchi

Table 3
Descriptive statistics and correlations ($N = 2854$).

Variables	Mean	SD	Min	Max	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
(1) Post-entry internationalization speed	7.39	0.86	0.01	8.25															
(2) Depth of international experience	6.87	1.64	1.10	9.64	-0.48														
(3) Breadth of international experience	0.24	0.20	0.02	1.00	-0.36	0.52													
(4) Big Data Analytics implementation	3.06	1.12	0.02	4.50	0.15	-0.26	-0.16												
(5) Financial leverage	0.39	0.22	0.00	1.00	-0.21	0.29	0.48	-0.02											
(6) Firm age	3.96	0.79	0.69	5.38	-0.09	0.55	0.38	-0.22	0.24										
(7) Firm size	3.47	1.15	0.05	6.36	-0.24	0.36	0.12	-0.44	0.06	0.25									
(8) ROA	0.08	0.09	-0.52	0.53	-0.02	0.05	0.09	0.05	-0.01	0.09	0.10								
(9) Board independence	0.86	0.14	0.14	1.00	-0.05	0.06	0.04	0.03	0.07	0.04	-0.01	0.03							
(10) Board size	27.69	12.07	3.00	118.00	0.02	0.23	0.11	-0.19	0.04	0.27	0.22	0.22	0.00						
(11) Institutional distance	1.12	0.53	-1.26	2.09	0.02	-0.24	-0.28	-0.07	-0.15	-0.07	-0.07	-0.07	-0.09	-0.07					
(12) Cultural distance	1.27	0.60	0.10	3.56	-0.02	-0.05	0.42	0.02	0.26	0.07	0.08	0.05	-0.01	-0.01	-0.37				
(13) Competitive intensity	0.94	0.05	0.84	0.98	-0.01	0.22	0.33	-0.05	0.27	0.26	-0.11	0.00	-0.09	0.16	-0.16	0.21			
(14) Innovation	5.16	3.43	0.00	11.08	-0.21	0.50	0.45	-0.29	0.33	0.33	0.26	0.02	-0.02	0.31	-0.16	0.20	0.48		
(15) Entrepreneurial orientation	0.09	0.10	0.00	1.00	-0.60	0.59	0.55	-0.12	0.37	0.33	0.23	-0.05	0.07	0.06	-0.21	0.21	0.18	0.36	
(16) CEO duality	0.20	0.40	0.00	1.00	0.02	0.15	0.12	-0.03	0.05	0.14	0.10	0.05	0.04	0.27	-0.01	0.02	0.09	0.20	0.09

Table 4
GMM results.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Depth of international experience		0.025*** [0.004] (0.000)	-0.017*** [0.002] (0.000)	-0.019** [0.006] (0.001)	-0.021*** [0.001] (0.000)	-0.031*** [0.003] (0.000)	-0.019*** [0.002] (0.000)	-0.018*** [0.002] (0.000)
Breadth of international experience		-0.584*** [0.058] (0.000)	-0.208*** [0.017] (0.000)	-0.458*** [0.074] (0.000)	-0.455*** [0.019] (0.000)	-0.365*** [0.032] (0.000)	-0.475*** [0.019] (0.000)	-0.352*** [0.024] (0.000)
Big Data Analytics implementation		-0.006 [0.004] (0.149)	-0.002* [0.001] (0.032)	-0.006 [0.005] (0.223)	-0.003** [0.001] (0.002)	-0.003 [0.003] (0.297)	-0.002 [0.002] (0.414)	-0.008*** [0.002] (0.000)
Depth of international experience squared			-0.003*** [0.000] (0.000)		-0.005*** [0.000] (0.000)	-0.005*** [0.001] (0.000)	-0.004*** [0.000] (0.000)	-0.002*** [0.000] (0.000)
Breadth of international experience squared				0.416* [0.180] (0.021)	0.699*** [0.054] (0.000)	0.640*** [0.058] (0.000)	0.497*** [0.054] (0.000)	0.529*** [0.052] (0.000)
Depth of international experience*Big Data Analytics implementation						0.005*** [0.001] (0.000)		0.017*** [0.001] (0.000)
Depth of international experience squared*Big Data Analytics implementation						0.004*** [0.000] (0.000)		0.006*** [0.000] (0.000)
Breadth of international experience*Big Data Analytics implementation							-0.003 [0.008] (0.682)	-0.024 [0.015] (0.107)
Breadth of international experience squared*Big Data Analytics implementation							-0.117*** [0.035] (0.001)	-0.185*** [0.041] (0.000)
Financial leverage	-0.053* [0.023] (0.022)	0.042** [0.014] (0.003)	0.036*** [0.004] (0.000)	0.078** [0.024] (0.001)	0.034*** [0.006] (0.000)	0.046*** [0.009] (0.000)	0.036*** [0.006] (0.000)	0.021* [0.008] (0.011)
Firm age	0.016* [0.008] (0.035)	0.030*** [0.006] (0.000)	0.028*** [0.002] (0.000)	0.057*** [0.011] (0.000)	0.033*** [0.003] (0.000)	0.054*** [0.004] (0.000)	0.031*** [0.003] (0.000)	0.046*** [0.003] (0.000)
Firm size	-0.012* [0.006] (0.035)	-0.028*** [0.004] (0.000)	0.013*** [0.001] (0.000)	0.012~ [0.007] (0.074)	0.017*** [0.001] (0.000)	0.017*** [0.002] (0.000)	0.013*** [0.001] (0.000)	0.012*** [0.002] (0.000)
ROA	-0.001** [0.000] (0.009)	-0.000~ [0.000] (0.095)	-0.002*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.003*** [0.000] (0.000)
Board independence	-0.033 [0.028] (0.242)	-0.023 [0.020] (0.268)	-0.017* [0.008] (0.022)	0.041 [0.030] (0.165)	-0.027*** [0.006] (0.000)	0.044*** [0.013] (0.001)	-0.018*** [0.005] (0.000)	0.048*** [0.012] (0.000)
Board size	0.000 [0.000] (0.947)	0.000 [0.000] (0.328)	0.000*** [0.000] (0.000)	-0.000 [0.000] (0.229)	0.000*** [0.000] (0.000)	0.001*** [0.000] (0.000)	0.000*** [0.000] (0.000)	0.001*** [0.000] (0.000)
Institutional distance	-0.013 [0.008] (0.121)	-0.007 [0.008] (0.351)	-0.007** [0.003] (0.010)	-0.039** [0.014] (0.006)	-0.003 [0.003] (0.282)	-0.015** [0.005] (0.005)	-0.002 [0.003] (0.480)	-0.012* [0.005] (0.033)
Cultural distance	0.010 [0.011] (0.333)	0.077*** [0.009] (0.000)	0.010*** [0.003] (0.000)	-0.005 [0.014] (0.739)	0.015*** [0.002] (0.000)	-0.002 [0.004] (0.621)	0.016*** [0.002] (0.000)	-0.005* [0.002] (0.011)
Competitive intensity	-0.066 [0.188] (0.727)	-0.037 [0.068] (0.585)	-0.082* [0.036] (0.023)	0.599* [0.303] (0.048)	-0.129* [0.062] (0.040)	-0.023 [0.056] (0.683)	-0.077~ [0.042] (0.070)	-0.129*** [0.039] (0.001)
Innovation	-0.003 [0.002] (0.202)	-0.003** [0.001] (0.008)	-0.011*** [0.001] (0.000)	-0.003 [0.002] (0.149)	-0.009*** [0.000] (0.000)	-0.006*** [0.001] (0.000)	-0.007*** [0.000] (0.000)	-0.006*** [0.001] (0.000)
Entrepreneurial orientation	-1.054*** [0.088] (0.000)	-0.714*** [0.068] (0.000)	-0.387*** [0.021] (0.000)	-0.461*** [0.089] (0.000)	-0.268*** [0.015] (0.000)	-0.479*** [0.040] (0.000)	-0.249*** [0.017] (0.000)	-0.529*** [0.032] (0.000)
CEO duality	0.017~ [0.009] (0.054)	0.030*** [0.006] (0.000)	0.015*** [0.002] (0.000)	0.018* [0.008] (0.026)	0.018*** [0.002] (0.000)	0.012*** [0.003] (0.000)	0.015*** [0.002] (0.000)	0.012*** [0.002] (0.000)
Lagged post-entry internationalization speed	0.787*** [0.003] (0.000)	0.793*** [0.002] (0.000)	0.871*** [0.001] (0.000)	0.834*** [0.003] (0.000)	0.872*** [0.001] (0.000)	0.841*** [0.001] (0.000)	0.871*** [0.002] (0.000)	0.850*** [0.001] (0.000)
Constant	1.828*** [0.170] (0.000)	1.524*** [0.074] (0.000)	0.976*** [0.010] (0.000)	1.238*** [0.032] (0.000)	0.930*** [0.010] (0.000)	1.185*** [0.014] (0.000)	0.934*** [0.012] (0.000)	1.113*** [0.012] (0.000)
Year dummies	Included	Included	Included	Included	Included	Included	Included	Included
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included
Number of observations	2983.000	2854.000	2854.000	2854.000	2854.000	2854.000	2854.000	2854.000
AR(1)	0.003	0.000	0.000	0.000	0.000	0.000	0.000	0.000
AR(2)	0.070	0.071	0.405	0.152	0.385	0.356	0.343	0.307
Hansen	22.720	108.191	206.187	87.385	200.286	184.412	191.657	198.337
P-value of Hansen J statistics	0.303	0.636	1.000	0.558	1.000	0.298	1.000	0.998

Notes: (1) $\sim p < 0.1$, $* p < 0.05$, $** p < 0.01$, $*** p < 0.001$ (two-tailed tests); (2) standard errors in parentheses; p-values in square brackets

Table 5

U-test result.

	Turning point (centered)	ICL	uCL	Within bounds	Lower bound: t(p)	Upper bound: t(p)	Overall t-statistics	Overall p-value
Depth of international experience	-2.05	-2.36	-1.77	Yes	10.63 (0.00)	-14.89 (0.00)	10.63	0.00
Breadth of international experience	0.33	0.30	0.37	Yes	-28.78 (0.00)	11.14 (0.00)	11.14	0.00

Turning point (centred) – estimated turning point in years. ICL and uCL – lower and upper 95% confidence limits for the turning point. Within bounds – indicates whether the turning point falls within the observed data ranges: depth of international experience (-5.79–3.12) and breadth (-0.17–0.79). The turning point is “within bounds” if both ICL and uCL lie inside these ranges.

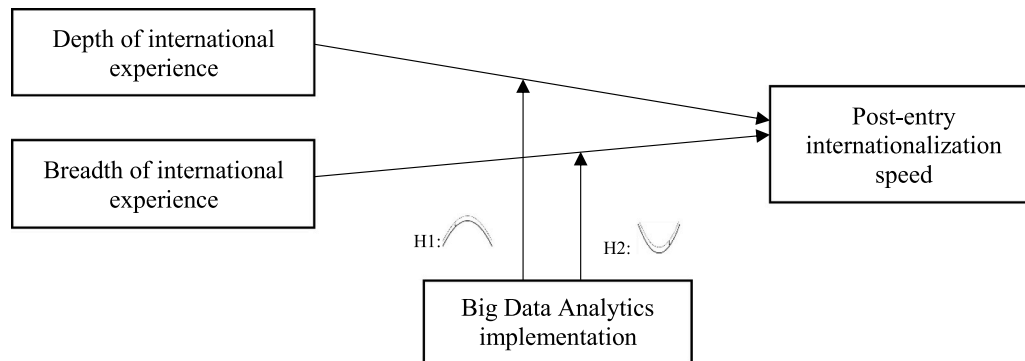


Fig. 1. Conceptual framework.

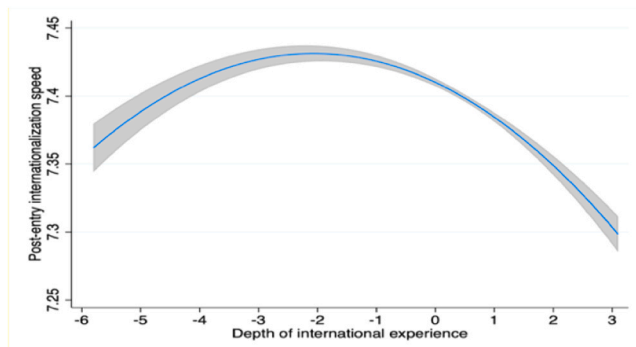


Fig. 2. The impact of depth of international experience on post-entry internationalization speed.

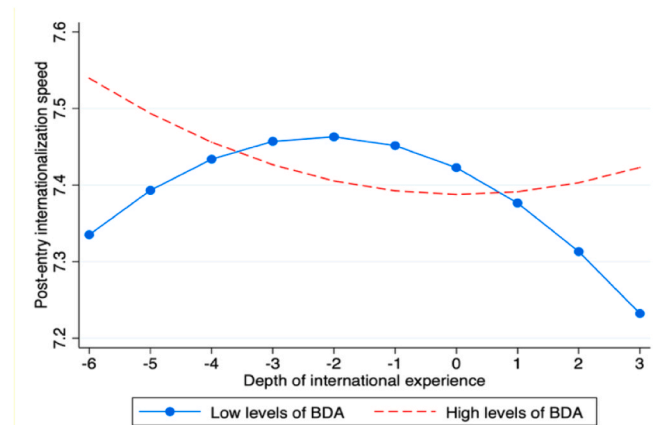


Fig. 4. Big Data Analytics as moderator in the depth - post-entry internationalization speed relationship.

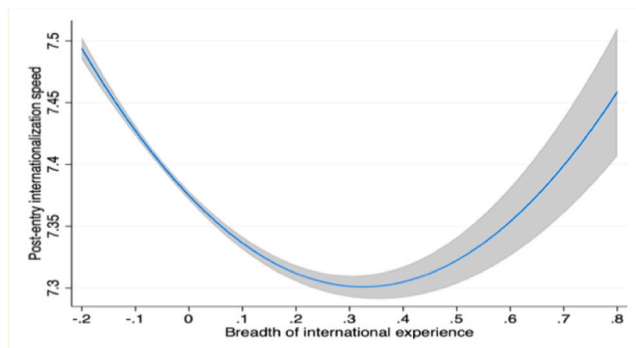


Fig. 3. The impact of breadth of international experience on post-entry internationalization speed.

test results (Table 5) and visual representations in Figs. 2 and 3 further

corroborate these shapes.

Models 6, 7, and 8 examine the conditioning role of Big Data Analytics implementation. H1 predicts that high BDA weakens the inverted U-shaped relationship for depth while H2 predicts that high BDA flattens the U-shaped relationship for breadth. Model 6 shows a significantly positive interaction between the squared depth and Big Data Analytics ($\beta = 0.004$, $p = 0.000$). The mean interaction effect from our marginal effect analysis is also significant ($\beta = 0.138$, $p = 0.000$). Model 7 tests the moderation for breadth (H2). The interaction between the squared breadth term and Big Data Analytics is negative and significant ($\beta = -0.117$, $p = 0.000$). Our marginal effect analysis confirms the significance of the mean interaction effect ($\beta = -0.235$, $p = 0.012$). Model 8 incorporates both moderating effects simultaneously. The results remain consistent, with the interaction term for depth ($\beta = 0.006$, $p = 0.000$) and breadth ($\beta = -0.185$, $p = 0.000$) retaining their significance and

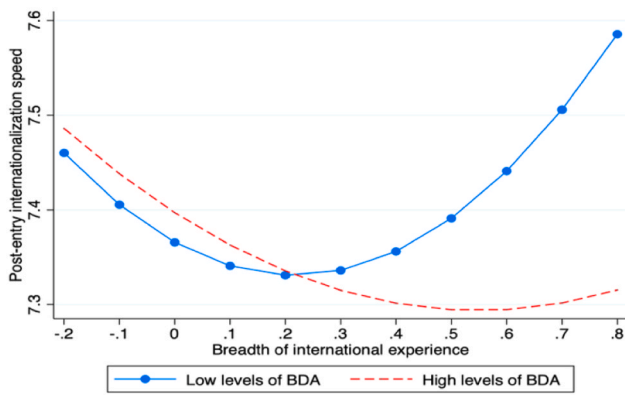


Fig. 5. Big Data Analytics as moderator in the breadth–post-entry internationalization speed relationship.

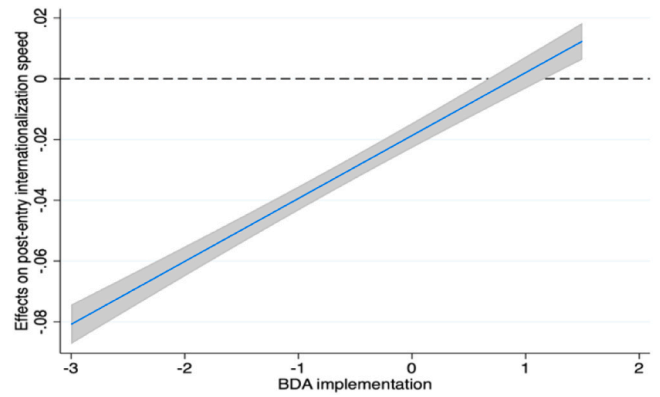


Fig. 8. Marginal effects of depth of international experience: Contingent on BDA implementation.

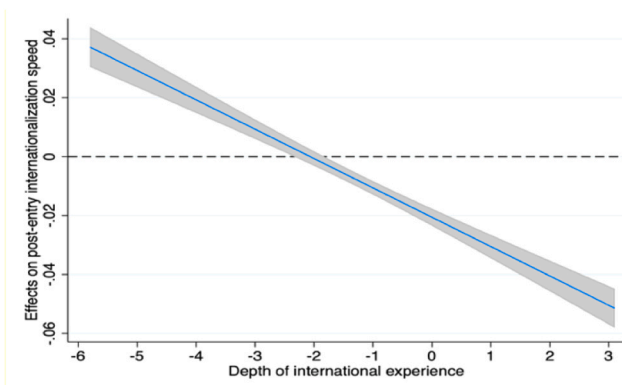


Fig. 6. Marginal effects of depth of international experience.

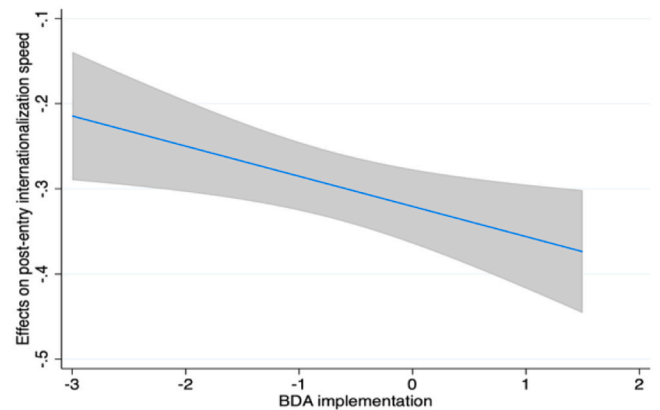


Fig. 9. Marginal effects of breadth of international experience: Contingent on BDA implementation.

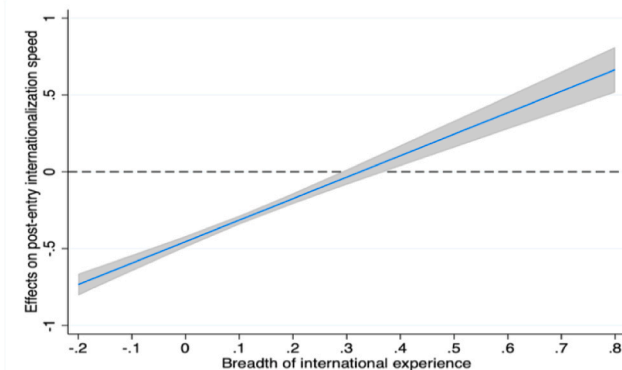


Fig. 7. Marginal effects of breadth of international experience.

direction. This confirms the robustness of the findings and visual representations in Figs. 4 and 5 further substantiating the validity of the moderating effect of Big Data Analytics implementation.

5.1.2. Marginal effects

Examining marginal effects, we find support for the hypothesized curvilinear relationships. Figs. 6–9 plot these effects (solid lines) with 95% confidence intervals (shaded areas) to identify regions of statistical significance. For depth of international experience (H0a), Fig. 6 shows three distinct regions: a significant positive effect at low (mean-centered) depth (between -5.8 and -2.4), representing the upward slope; a non-significant transition region around the peak (between -2.4

and -1.9); and a significant negative effect at high depth (above -1.9), representing the downward slope. In terms of economic significance, post-entry internationalization speed increases by 6.7% as depth rises to the peak, and decreases by 5.2% as depth moves one standard deviation beyond the peak. For breadth of international experience (H0b), Fig. 7 reveals three distinct regions: a significant negative effect at low (mean-centered) breadth (between -0.2 and 0.2), representing the downward slope before the trough; a non-significant transition region at the trough (between 0.2 and 0.3); and a significant positive effect at high breadth (above 0.3), representing the upward slope. In terms of economic significance, post-entry internationalization speed decreases by 1.3% as breadth approaches the turning point from one standard deviation below and increases by 1.3% as it moves one standard deviation beyond the turning point.

Turning to the moderating role of BDA on depth (H1), Fig. 8 displays the marginal effect of depth on post-entry internationalization speed across the full range of BDA implementation. At low (mean-centered) BDA levels (from -3.1 – 0.5), the marginal effect of the quadratic depth term is significantly negative. This effect becomes statistically insignificant between approximately 0.5 and 0.9 and turns significantly positive above 0.9 . Thus, higher BDA implementation reverses the curvature of the depth–speed relationship, supporting H1. For the moderation of BDA on breadth (H2), Fig. 9 reveals a different pattern than for depth: across the full observed range of BDA (-3.1 – 1.2), the marginal effect of the quadratic breadth term remains negative and statistically significant. While the effect never crosses into non-significance or reverses sign, the U-shaped curvature diminishes monotonically as BDA increases. This consistent flattening of the relationship provides strong empirical support for H2.

Table 6

Regression results (alternative measures of Big Data Analytics).

	Alternative measure of Big Data Analytics (CATA)				Alternative measure of Big Data Analytics (binary)			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Depth of international experience	-0.019*** [0.001] (0.000)	-0.104*** [0.017] (0.000)	-0.021*** [0.001] (0.000)	-0.147*** [0.013] (0.000)	-0.025*** [0.001] (0.000)	-0.027*** [0.002] (0.000)	-0.015* [0.007] (0.029)	-0.038*** [0.002] (0.000)
Breadth of international experience	-0.389*** [0.017] (0.000)	-0.397*** [0.031] (0.000)	-1.016*** [0.100] (0.000)	-0.500*** [0.138] (0.000)	-0.266*** [0.015] (0.000)	-0.316*** [0.020] (0.000)	-0.306*** [0.071] (0.000)	-0.213*** [0.019] (0.000)
Big Data Analytics implementation	0.066*** [0.014] (0.000)	-0.187*** [0.047] (0.000)	0.090*** [0.020] (0.000)	-0.112*** [0.022] (0.000)	0.005*** [0.001] (0.000)	-0.037*** [0.003] (0.000)	0.014 [0.013] (0.277)	-0.008~ [0.004] (0.075)
Depth of international experience squared	-0.003*** [0.000] (0.000)	-0.066*** [0.009] (0.000)	-0.004*** [0.000] (0.000)	-0.060*** [0.007] (0.000)	-0.003*** [0.000] (0.000)	-0.007*** [0.000] (0.000)	0.004** [0.001] (0.008)	-0.008*** [0.000] (0.000)
Breadth of international experience squared	0.520*** [0.040] (0.000)	0.560*** [0.072] (0.000)	1.877*** [0.391] (0.000)	2.899*** [0.434] (0.000)	0.443*** [0.043] (0.000)	0.410*** [0.045] (0.000)	0.341* [0.166] (0.039)	0.692*** [0.052] (0.000)
Depth of international experience*Big Data Analytics implementation		0.084*** [0.017] (0.000)		0.130*** [0.013] (0.000)		0.019*** [0.002] (0.000)		0.028*** [0.003] (0.000)
Depth of international experience squared*Big Data Analytics implementation		0.067*** [0.010] (0.000)		0.060*** [0.007] (0.000)		0.010*** [0.001] (0.000)		0.011*** [0.001] (0.000)
Breadth of international experience*Big Data Analytics implementation			0.674*** [0.105] (0.000)	0.267~ [0.144] (0.063)			0.197** [0.062] (0.001)	-0.030 [0.028] (0.273)
Breadth of international experience squared*Big Data Analytics implementation			-1.438*** [0.395] (0.000)	-2.724*** [0.452] (0.000)			-0.642~ [0.328] (0.051)	-0.784*** [0.127] (0.000)
Financial leverage	0.028*** [0.005] (0.000)	0.071*** [0.009] (0.000)	0.022*** [0.005] (0.000)	0.019*** [0.005] (0.000)	0.017** [0.005] (0.001)	0.007 [0.006] (0.263)	0.103*** [0.024] (0.000)	0.048*** [0.007] (0.000)
Firm age	0.029*** [0.002] (0.000)	0.053*** [0.004] (0.000)	0.032*** [0.003] (0.000)	0.038*** [0.003] (0.000)	0.034*** [0.003] (0.000)	0.037*** [0.002] (0.000)	0.045*** [0.009] (0.000)	0.040*** [0.003] (0.000)
Firm size	0.019*** [0.001] (0.000)	0.014*** [0.002] (0.000)	0.019*** [0.001] (0.000)	0.017*** [0.001] (0.000)	0.017*** [0.001] (0.000)	0.015*** [0.002] (0.000)	0.013* [0.005] (0.012)	0.021*** [0.001] (0.000)
ROA	-0.002*** [0.000] (0.000)	-0.003*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.001*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.003*** [0.000] (0.000)	-0.003*** [0.000] (0.000)	-0.004*** [0.000] (0.000)
Board independence	0.004 [0.005] (0.387)	0.044*** [0.011] (0.000)	-0.005 [0.005] (0.330)	-0.015** [0.005] (0.006)	0.019*** [0.005] (0.000)	-0.009 [0.006] (0.166)	-0.011 [0.026] (0.670)	0.022** [0.007] (0.001)
Board size	0.000*** [0.000] (0.000)	0.000** [0.000] (0.005)	0.000*** [0.000] (0.000)	0.000** [0.000] (0.001)	0.000*** [0.000] (0.000)	0.000 [0.000] (0.905)	-0.000 [0.000] (0.151)	0.000~ [0.000] (0.057)
Institutional distance	0.005* [0.003] (0.043)	0.005 [0.006] (0.371)	0.001 [0.003] (0.815)	-0.001 [0.003] (0.623)	-0.004~ [0.002] (0.075)	-0.007** [0.003] (0.009)	-0.023~ [0.013] (0.091)	-0.020*** [0.003] (0.000)
Cultural distance	0.005* [0.002] (0.011)	-0.016*** [0.004] (0.000)	0.008*** [0.002] (0.000)	-0.001 [0.002] (0.806)	-0.010*** [0.002] (0.000)	-0.012*** [0.003] (0.000)	-0.054*** [0.012] (0.000)	-0.024*** [0.003] (0.000)
Competitive intensity	0.023 [0.058] (0.693)	0.155** [0.059] (0.008)	-0.034 [0.037] (0.361)	-0.055* [0.028] (0.049)	-0.174** [0.054] (0.001)	-0.006 [0.032] (0.856)	0.544** [0.178] (0.002)	0.244*** [0.039] (0.000)
Innovation	-0.007*** [0.000] (0.000)	-0.006*** [0.001] (0.000)	-0.007*** [0.000] (0.000)	-0.005*** [0.000] (0.000)	-0.009*** [0.000] (0.000)	-0.010*** [0.000] (0.000)	-0.005** [0.002] (0.005)	-0.008*** [0.001] (0.000)
Entrepreneurial orientation	-0.305*** [0.016] (0.000)	-0.437*** [0.043] (0.000)	-0.233*** [0.019] (0.000)	-0.346*** [0.018] (0.000)	-0.275*** [0.018] (0.000)	-0.317*** [0.028] (0.000)	-0.637*** [0.116] (0.000)	-0.414*** [0.019] (0.000)
CEO duality	0.016*** [0.001] (0.000)	0.013*** [0.003] (0.000)	0.012*** [0.002] (0.000)	0.012*** [0.002] (0.000)	0.016*** [0.001] (0.000)	0.016*** [0.002] (0.000)	0.023*** [0.007] (0.001)	0.010*** [0.003] (0.001)
Lagged post-entry internationalization speed	0.879*** [0.001] (0.000)	0.846*** [0.001] (0.000)	0.881*** [0.001] (0.000)	0.886*** [0.001] (0.000)	0.881*** [0.001] (0.000)	0.866*** [0.001] (0.000)	0.837*** [0.003] (0.000)	0.864*** [0.001] (0.000)
Constant	0.815*** [0.017] (0.000)	1.288*** [0.048] (0.000)	0.787*** [0.025] (0.000)	0.963*** [0.022] (0.000)	0.879*** [0.008] (0.000)	1.016*** [0.008] (0.000)	1.211*** [0.026] (0.000)	1.026*** [0.010] (0.000)
Year dummies	Included	Included	Included	Included	Included	Included	Included	Included
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included
Number of observations	2882.000	2882.000	2882.000	2882.000	2981.000	2981.000	2981.000	2981.000
AR(1)	0.000	0.000	0.001	0.001	0.000	0.000	0.000	0.000

(continued on next page)

Table 6 (continued)

	Alternative measure of Big Data Analytics (CATA)				Alternative measure of Big Data Analytics (binary)			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
AR(2)	0.614	0.274	0.638	0.456	0.506	0.287	0.268	0.262
Hansen	210.666	189.433	197.675	207.888	225.228	203.154	101.845	211.121
P-value of Hansen J statistics	1.000	0.186	1.000	1.000	1.000	1.000	0.514	0.991

Notes: (1) $\sim p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed tests); (2) standard errors in parentheses; p-values in square brackets

5.1.3. Additional analysis: the effect of BDA implementation

Model 8 reveals a statistically significant, albeit small, direct effect of Big Data Analytics on the post-entry internationalization speed ($\beta = -0.008$, $p = 0.000$). Because we log-transform cumulative Big Data Analytics announcements to address skewness, the coefficient captures proportional changes. Specifically, a 100% increase in Big Data Analytics level corresponds to a 0.005-unit decrease in post-entry speed ($-0.008 \times \ln(2)$) ($\ln(2)$ is the change in the natural logarithm of Big Data Analytics when Big Data Analytics doubles). Relative to the sample mean speed of 7.38, this represents an approximately 0.05% reduction ($(-0.005 \times \ln(2) / 7.38) \times 100$). This modest negative effect likely reflects the early-stage frictions associated with Big Data Analytics adoption, such as upfront investments in IT infrastructure, data integration, and talent acquisition. Crucially, the strategic value of Big Data Analytics does not lie in this small direct effect but in its strong and significant moderating influence. By conditioning the curvilinear returns to depth and breadth, Big Data Analytics alters when, and how strongly, international experience accelerates post-entry internationalization. Hence, Big Data Analytics is best understood as a costly but transformative capability whose primary impact emerges through its indirect effects on experiential learning.

5.2. Robustness checks

To ensure robustness, we conduct several tests, and results remain fairly consistent across models.

5.2.1. Alternative measures of key variables

First, we use an alternative measure of depth of international experience, proxied by the total years a firm has operated abroad (Autio et al., 2000; Chao & Kumar, 2010). Second, we replace the multinational dependent variable with a flow measure of post-entry internationalization speed calculated as the year-on-year difference in post-entry internationalization speed to replicate the GMM model. Third, we employ two distinct measures of Big Data Analytics implementation: (1) a machine learning-based proportion of Big Data Analytics-related words in implementation from annual reports (scaled 0–1) (Bhandari et al., 2023). Full results are reported in Table 6 and Table 7.

5.2.2. Sample selection bias

To address potential sample selection bias stemming from the non-random decision of firms to (1) engage in internationalization and (2) implement Big Data Analytics, we employ the Heckman two-stage estimation procedure (Certo et al., 2016; Liu et al., 2023). In the first stage, we run two separate probit models: Model A predicts the likelihood of a firm's decision to internationalize; Model B predicts the likelihood of a firm's implementation of BDA (more detailed results for the first stage can be found in the Appendix). Both models include relevant firm-level resources and contextual covariates. To satisfy the exclusion restriction and ensure model identification, we introduce the instrumental variables into the first-stage equations variables that are theoretically related to the selection process but are excluded from the second-stage equation: product diversification and industry-level degree of internationalization (enters first-stage Model A only), and industry-level BDA implementation intensity (enters first-stage Model B only). The inverse Mills ratios from this stage are then incorporated in the second stage, estimated using the two-step system GMM (Blundell &

Bond, 2000). The findings are reported in Table 8, and they remain consistent.

5.2.3. Alternative samples

We also re-run regressions using pre-Covid data (2010–2019; Appendix D) to eliminate pandemic effects.

5.2.4. Untangling the selection effect and nescience effect

To better understand the moderating role of Big Data Analytics, we provide both conceptual clarification and empirical evidence showing that these two mechanisms manifest differently in the Big Data Analytics - international experience interaction. Consistent with Balasubramanian et al. (2022), we view selection and nescience effects as complementary learning dynamics that jointly determine how firms navigate the complex interplay between experiential and data-driven learning.

The selection effect emerges when Big Data Analytics-driven patterning converges with experiential cues to narrow the set of viable actions, which manifests empirically as a change in the steepness of nonlinear experience - post-entry internationalization speed relationships (i.e., reduced or altered curvature). In terms of the nescience effect, the interaction between data-driven prescriptions and incomplete experiential knowledge shifts managers away from the contextually strategic level under low level of Big Data Analytics, which empirically appears as a relocation of the optimal strategic level (the experience intensity at which post-entry internationalization speed is maximized or where marginal benefits turn positive).

Table 9 demonstrates clear and separate empirical signatures of selection and nescience effects. For depth of international experience, the magnitude of the inverted U-shaped curvature changes significantly (from -0.005 to -0.003), resulting in a total magnitude change of $= 0.0083$. This flattening of the inverted U-shape indicates that under high Big Data Analytics, the post-entry internationalization speed of firms becomes substantially less sensitive to having too little or too much international experience. For breadth of international experience, the U-shaped curvature is reduced by 54% under high Big Data Analytics (from 0.545 to 0.254). This dramatic reduction shows that Big Data Analytics' logic powerfully narrows the range of achievable post-entry internationalization speed outcomes across different experiential portfolios. The consistent, significant reduction in curvature magnitude operationalizes the core of the selection effect: Big Data Analytics acts as a variance-reducing moderator, making the rate of post-entry internationalization speed less dependent on a firm's level of international experience. The nescience effect is operationalized by the significant relocation of the turning point under high Big Data Analytics. For depth, the turning point shifts from -2.48 to -0.27 . For breadth of international experience, the inflection point shifts from 0.22 to 0.81. These changes show that Big Data Analytics redefines where experiential learning yields the greatest net benefit.

6. Discussion

6.1. Contributions to theory

6.1.1. For organizational learning research

We advance organizational learning theory and, importantly, the Uppsala model of internationalization by theorizing how digital technologies, specifically Big Data Analytics, reconfigure the learning

Table 7

Regression results (Alternative measure of post-entry internationalization speed and depth of international experience).

	Alternative measure of post-entry internationalization speed				Alternative measure of depth of international experience			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Depth of international experience	-0.003*** [0.000] (0.000)	0.001* [0.001] (0.031)	-0.012*** [0.003] (0.000)	-0.006* [0.003] (0.020)	0.069*** [0.003] (0.000)	0.066*** [0.003] (0.000)	0.107*** [0.003] (0.000)	0.118*** [0.011] (0.000)
Breadth of international experience	0.033*** [0.006] (0.000)	-0.018** [0.006] (0.002)	0.072 [0.047] (0.123)	0.048 [0.036] (0.178)	-0.402*** [0.020] (0.000)	-0.311*** [0.020] (0.000)	-0.337*** [0.022] (0.000)	-0.211* [0.090] (0.019)
Big Data Analytics implementation	-0.002*** [0.000] (0.000)	-0.006*** [0.000] (0.000)	0.002 [0.003] (0.511)	0.002 [0.003] (0.487)	-0.004*** [0.001] (0.000)	-0.000 [0.002] (0.884)	-0.011*** [0.002] (0.000)	-0.011 [0.008] (0.142)
Depth of international experience squared	-0.002*** [0.000] (0.000)	-0.000 [0.000] (0.588)	-0.002** [0.001] (0.001)	0.001 [0.001] (0.182)	0.045*** [0.001] (0.000)	0.044*** [0.001] (0.000)	0.056*** [0.001] (0.000)	0.053*** [0.005] (0.000)
Breadth of international experience squared	0.036~ [0.019] (0.055)	0.070*** [0.016] (0.000)	0.022 [0.079] (0.783)	-0.019 [0.074] (0.796)	0.560*** [0.054] (0.000)	0.343*** [0.049] (0.000)	0.471*** [0.059] (0.000)	0.491* [0.192] (0.011)
Depth of international experience*Big Data Analytics implementation		0.001*** [0.000] (0.000)		0.005*** [0.001] (0.000)		0.009*** [0.002] (0.000)		0.004 [0.008] (0.622)
Depth of international experience squared*Big Data Analytics implementation		0.000** [0.000] (0.002)		-0.001 [0.000] (0.207)		-0.010*** [0.001] (0.000)		-0.008 [0.006] (0.204)
Breadth of international experience*Big Data Analytics implementation			0.066*** [0.015] (0.000)	0.016 [0.016] (0.339)			0.050*** [0.009] (0.000)	-0.067 [0.050] (0.179)
Breadth of international experience squared*Big Data Analytics implementation			-0.195*** [0.051] (0.000)	-0.069 [0.055] (0.208)			-0.089** [0.032] (0.005)	0.259 [0.159] (0.105)
Financial leverage	0.017*** [0.002] (0.000)	0.015*** [0.002] (0.000)	-0.013 [0.012] (0.302)	0.023* [0.010] (0.022)	0.001 [0.006] (0.924)	-0.010* [0.005] (0.045)	0.042*** [0.009] (0.000)	-0.058* [0.026] (0.025)
Firm age	0.008*** [0.001] (0.000)	0.001 [0.001] (0.164)	0.009* [0.004] (0.036)	0.016*** [0.003] (0.000)	0.009*** [0.002] (0.000)	0.008** [0.002] (0.001)	0.012*** [0.003] (0.000)	0.004 [0.007] (0.630)
Firm size	0.002** [0.001] (0.008)	-0.002*** [0.001] (0.001)	0.005~ [0.052] (0.052)	0.004~ [0.002] (0.093)	-0.003** [0.001] (0.001)	0.000 [0.001] (0.961)	-0.023*** [0.001] (0.000)	-0.048*** [0.006] (0.000)
ROA	-0.000*** [0.000] (0.000)	0.000*** [0.000] (0.000)	-0.000~ [0.000] (0.053)	0.000 [0.000] (0.230)	-0.002*** [0.000] (0.000)	-0.001*** [0.000] (0.000)	-0.003*** [0.000] (0.000)	-0.001* [0.001] (0.047)
Board independence	-0.006* [0.003] (0.046)	0.006* [0.003] (0.032)	0.065*** [0.013] (0.000)	0.095*** [0.014] (0.000)	-0.010* [0.004] (0.016)	-0.011~ [0.006] (0.066)	-0.029* [0.012] (0.021)	-0.106** [0.039] (0.006)
Board size	-0.000*** [0.000] (0.000)	-0.000*** [0.000] (0.000)	-0.001** [0.000] (0.001)	-0.000 [0.000] (0.125)	0.000* [0.000] (0.025)	-0.000 [0.000] (0.604)	0.000 [0.000] (0.450)	-0.000 [0.000] (0.848)
Institutional distance	0.008*** [0.001] (0.000)	0.003* [0.001] (0.016)	0.011 [0.007] (0.115)	0.009~ [0.005] (0.086)	0.011*** [0.003] (0.001)	0.012*** [0.003] (0.000)	0.002 [0.004] (0.696)	0.006 [0.013] (0.632)
Cultural distance	-0.021*** [0.001] (0.000)	-0.009*** [0.001] (0.000)	-0.054*** [0.005] (0.000)	-0.043*** [0.004] (0.000)	0.024*** [0.003] (0.000)	0.006~ [0.003] (0.072)	0.017*** [0.002] (0.000)	0.017 [0.011] (0.118)
Competitive intensity	-0.013 [0.027] (0.633)	-0.093*** [0.016] (0.000)	0.285* [0.123] (0.020)	0.174 [0.112] (0.122)	-0.021 [0.029] (0.476)	-0.118** [0.041] (0.004)	0.311*** [0.051] (0.000)	0.060 [0.243] (0.804)
Innovation	-0.002*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.003** [0.001] (0.003)	-0.001 [0.001] (0.128)	-0.012*** [0.001] (0.000)	-0.011*** [0.001] (0.000)	-0.009*** [0.001] (0.000)	-0.003 [0.003] (0.201)
Entrepreneurial orientation	-0.043*** [0.007] (0.000)	-0.016** [0.005] (0.001)	0.068 [0.055] (0.216)	-0.005 [0.043] (0.904)	-0.510*** [0.019] (0.000)	-0.452*** [0.018] (0.000)	-0.699*** [0.032] (0.000)	-1.050*** [0.107] (0.000)
CEO duality	0.002** [0.001] (0.001)	0.003*** [0.001] (0.000)	0.005* [0.002] (0.037)	0.001 [0.002] (0.558)	0.017*** [0.002] (0.000)	0.010*** [0.002] (0.000)	0.015*** [0.002] (0.000)	0.020** [0.008] (0.009)
Lagged post-entry internationalization speed	0.489*** [0.000] (0.000)	0.502*** [0.001] (0.000)	0.488*** [0.001] (0.000)	0.494*** [0.001] (0.000)	0.865*** [0.002] (0.000)	0.875*** [0.002] (0.000)	0.847*** [0.002] (0.000)	0.801*** [0.007] (0.000)
Constant	0.003 [0.002] (0.142)	-0.008*** [0.002] (0.000)	0.010 [0.011] (0.362)	-0.002 [0.009] (0.852)	0.934*** [0.011] (0.000)	0.877*** [0.019] (0.000)	1.063*** [0.011] (0.000)	1.404*** [0.051] (0.000)
Year dummies	Included	Included	Included	Included	Included	Included	Included	Included
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included
Number of observations	2663.000	2663.000	2663.000	2663.000	2854.000	2854.000	2854.000	2854.000

(continued on next page)

Table 7 (continued)

	Alternative measure of post-entry internationalization speed				Alternative measure of depth of international experience			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
AR(1)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
AR(2)	0.060	0.069	0.084	0.077	0.206	0.182	0.205	0.252
Hansen	210.465	202.329	100.100	103.713	204.928	195.178	208.509	84.042
P-value of Hansen J statistics	0.983	1.000	0.394	0.462	1.000	1.000	1.000	0.684

Notes: (1) $\sim p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed tests); (2) standard errors in parentheses; p-values in square brackets

mechanisms underpinning international experience. Our study reveals how algorithmic learning through Big Data Analytics disrupts this trajectory by introducing a competing algorithmic learning logic (Balasubramanian et al., 2022), which challenges the assumptions that learning is path-dependent and locally embedded, that routines evolve through selective experiential retention, and that errors and anomalies serve as essential catalysts for learning. By integrating these tensions with our empirical findings, we demonstrate that Big Data Analytics does not merely alter the strength of experiential learning effects but fundamentally restructures where, how, and under what conditions the core mechanisms of experiential learning operate across different types of international experience.

First, we challenge the premise that learning is path-dependent and locally embedded. In traditional organization learning theory, knowledge develops through situated, context-specific interactions (Feldman & Pentland, 2003), which is also further reflected in the Uppsala model positing that firms reduce uncertainty through gradual, stepwise commitment, learning from direct market experience (Johanson & Vahne, 1977, 2009). Big Data Analytics' logic of formal rationality disrupts this by enabling decontextualized, algorithmic inferences. For depth, Big Data Analytics undermines the early formation of tacit, locally grounded routines, yet at higher levels of experience it also releases firms from the inertia of path dependence by surfacing non-obvious patterns that entrenched expertise may obscure. For breadth, Big Data Analytics facilitates rapid cross-contextual pattern recognition at low-to-moderate breadth, but this same generalization risks imposing a unitary interpretive logic on diverse national settings, diminishing the value of locally embedded insights.

Second, we re-tune the assumption that routines evolve through selective retention shaped by experiential feedback (March, 1991). In the Uppsala model, this also reflects Uppsala's change mechanism - the process by which market-specific experiential knowledge is converted into commitment decisions. Big Data Analytics shifts this evaluation process toward automated statistical optimization and decouples knowledge from direct experience. In breadth-based learning, Big Data Analytics accelerates early-stage knowledge transfer by identifying commonalities across markets, but eventually it induces algorithmic over-convergence that erodes contextual nuance and suppresses the benefits of a diversified experiential portfolio, undermining the Uppsala premise that a portfolio of market experiences enhances adaptive capability. In depth-based learning, Big Data Analytics can truncate early-stage trial-and-error of early learning but for firms with high depth, Big Data Analytics' formal rationality offers an external, de-biased perspective that can counteract the over-embeddedness of deep market knowledge, allowing firms to see beyond their own experiential path. This means the liability of foreignness is no longer countered solely by accumulated experience but can be pre-emptively reduced by algorithmic foresight.

Third, we question the assumption that anomalies and errors are central triggers for learning (Levitt & March, 1988). Big Data Analytics' emphasis on predictive stability leads to the suppression of irregularities, which directly conflicts with the anomaly-driven refinement central to experiential learning. For breadth, Big Data Analytics' filtering of contextual irregularities deprives firms of disconfirming signals drawn from diverse environments, which explains the diminished value of high

breadth in our results; for depth, it removes important local deviations that normally prompt adaptation at lower levels of depth, while paradoxically also helping correct biases rooted in overly localized past failures.

In sum, by linking these empirical insights back to the theoretical tensions articulated in the introduction, our study offers an extension of organizational learning theory. This also offers a nuanced account of digital-era learning and answers recent calls to explore technology's role in organizational learning (Argote et al., 2021).

6.1.2. For the Big Data Analytics literature

This study advances this literature by framing it not merely as a performance-enhancing capability, but as an interpretive moderator that reshapes how firms leverage experiential knowledge in IB contexts. While prior work highlights Big Data Analytics' performance benefits (Ghasemaghaei & Calic, 2019; Huang et al., 2020; Wamba et al., 2017), it has largely overlooked how organizations balance data-driven insights with experiential learning in uncertain environments. Our mechanism-based analysis explains the contingent value of Big Data Analytics: its selection effect can either efficiently guide or prematurely constrain strategic choice, while its nescience effect can either correct bias or create critical blind spots, depending on the maturity and type of experience. Moreover, by distinguishing Big Data Analytics' influence across experiential depth and breadth, we also respond to calls for finer-grained theorization of when algorithmic tools complement or conflict with human learning (Argote et al., 2021; Balasubramanian et al., 2022), thus providing a nuanced viewpoint on the harmonization of these knowledge streams to shape post-entry internationalization speed.

6.1.3. For internationalization speed studies

We contribute to the growing literature on post-entry internationalization speed by clarifying how firms' experiential knowledge foundations interact with digital technologies to accelerate or decelerate foreign expansion. We show that Big Data Analytics does not simply enhance or diminish the value of international experience but reallocates the mechanisms through which experiential knowledge (depth and breadth) contributes to post-entry internationalization speed. For breadth, Big Data Analytics reduces early-stage learning constraints at low-to-moderate levels but suppresses responsiveness at high levels through algorithmic overgeneralization. For depth, it fosters adaptability at high levels by countering inertia but may displace valuable learning-by-doing at moderate levels. In doing so, we move beyond deterministic narratives of technology-driven acceleration or experience-driven adaptation and instead position speed as an outcome shaped by the evolving interplay between algorithmic and experiential learning.

6.2. Implications for practice

Our findings offer practical insights for managers. First, Big Data Analytics should not be treated as a one-size-fits-all solution. Instead, managers must consider the strategic balance between algorithmic insights and experiential judgment. To operationalize this balance, firms should establish decision governance frameworks that clarify which

Table 8
Heckman second stage result.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Depth of international experience		0.027*** [0.005] (0.000)	-0.018*** [0.002] (0.000)	-0.023** [0.007] (0.001)	-0.030*** [0.004] (0.000)	-0.017*** [0.002] (0.000)	-0.021*** [0.002] (0.000)	-0.020*** [0.002] (0.000)
Breadth of international experience		-0.590*** [0.073] (0.000)	-0.247*** [0.014] (0.000)	-0.431*** [0.088] (0.000)	-0.356*** [0.053] (0.000)	-0.503*** [0.023] (0.000)	-0.485*** [0.019] (0.000)	-0.410*** [0.021] (0.000)
Big Data Analytics implementation		-0.007 [0.005] (0.123)	-0.005*** [0.001] (0.000)	-0.003 [0.006] (0.600)	0.000 [0.003] (0.881)	-0.015*** [0.002] (0.000)	-0.001 [0.002] (0.423)	-0.012*** [0.002] (0.000)
Depth of international experience squared			-0.003*** [0.000] (0.000)		-0.003** [0.001] (0.002)	-0.004*** [0.000] (0.000)	-0.005*** [0.000] (0.000)	-0.004*** [0.000] (0.000)
Breadth of international experience squared				0.514* [0.203] (0.011)	0.598*** [0.109] (0.000)	0.826*** [0.059] (0.000)	0.772*** [0.054] (0.000)	0.586*** [0.049] (0.000)
Depth of international experience*Big Data Analytics implementation						0.008*** [0.001] (0.000)		0.016*** [0.001] (0.000)
Depth of international experience squared*Big Data Analytics implementation						0.005*** [0.000] (0.000)		0.007*** [0.000] (0.000)
Breadth of international experience*Big Data Analytics implementation							-0.009 [0.010] (0.369)	-0.028* [0.013] (0.036)
Breadth of international experience squared*Big Data Analytics implementation							-0.105* [0.042] (0.012)	-0.225*** [0.041] (0.000)
Financial leverage	-0.060* [0.026] (0.023)	0.051** [0.018] (0.004)	0.047*** [0.006] (0.000)	0.059~ [0.031] (0.058)	0.037* [0.017] (0.031)	0.010 [0.008] (0.245)	0.019** [0.007] (0.004)	0.031*** [0.008] (0.000)
Firm age	0.014 [0.009] (0.111)	0.031*** [0.008] (0.000)	0.030*** [0.003] (0.000)	0.041** [0.014] (0.002)	0.052*** [0.006] (0.000)	0.052*** [0.004] (0.000)	0.031*** [0.003] (0.000)	0.052*** [0.003] (0.000)
Firm size	-0.014* [0.006] (0.024)	-0.032*** [0.005] (0.000)	0.019*** [0.002] (0.000)	0.010 [0.009] (0.264)	0.021*** [0.004] (0.000)	0.015*** [0.002] (0.000)	0.017*** [0.002] (0.000)	0.021*** [0.002] (0.000)
ROA	-0.001~ [0.000] (0.052)	-0.000 [0.000] (0.414)	-0.003*** [0.000] (0.000)	-0.002*** [0.001] (0.000)	-0.003*** [0.000] (0.000)	-0.004*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.004*** [0.000] (0.000)
Board independence	-0.019 [0.034] (0.570)	-0.049~ [0.025] (0.050)	-0.027*** [0.007] (0.000)	0.027 [0.042] (0.526)	0.111*** [0.020] (0.000)	-0.028** [0.011] (0.009)	-0.008 [0.007] (0.252)	0.045*** [0.011] (0.000)
Board size	0.000 [0.000] (0.470)	0.000 [0.000] (0.277)	0.000*** [0.000] (0.000)	-0.000 [0.000] (0.903)	0.001* [0.000] (0.020)	0.000 [0.000] (0.213)	0.000*** [0.000] (0.000)	0.001*** [0.000] (0.000)
Institutional distance	-0.014 [0.009] (0.132)	-0.004 [0.009] (0.701)	-0.005~ [0.003] (0.064)	-0.048** [0.017] (0.005)	-0.007 [0.008] (0.415)	-0.010* [0.005] (0.033)	-0.009* [0.004] (0.021)	-0.016*** [0.005] (0.000)
Cultural distance	0.016 [0.011] (0.174)	0.086*** [0.011] (0.000)	0.011*** [0.002] (0.000)	-0.008 [0.018] (0.641)	-0.003 [0.008] (0.689)	0.019*** [0.002] (0.000)	0.020*** [0.002] (0.000)	0.006* [0.003] (0.043)
Competitive intensity	-0.148 [0.216] (0.494)	-0.068 [0.087] (0.434)	-0.067 [0.048] (0.155)	0.206 [0.402] (0.609)	0.576** [0.180] (0.001)	-0.041 [0.035] (0.244)	-0.125** [0.040] (0.002)	0.033 [0.046] (0.477)
Innovation	-0.004 [0.003] (0.150)	-0.003~ [0.002] (0.076)	-0.013*** [0.001] (0.000)	-0.000 [0.003] (0.958)	-0.008*** [0.001] (0.000)	-0.008*** [0.001] (0.000)	-0.009*** [0.001] (0.000)	-0.008*** [0.001] (0.000)
Entrepreneurial orientation	-0.950*** [0.121] (0.000)	-0.752*** [0.091] (0.000)	-0.382*** [0.021] (0.000)	-0.466*** [0.108] (0.000)	-0.412*** [0.049] (0.000)	-0.470*** [0.031] (0.000)	-0.279*** [0.023] (0.000)	-0.514*** [0.039] (0.000)
CEO duality	0.023* [0.009] (0.015)	0.027*** [0.007] (0.000)	0.016*** [0.002] (0.000)	0.017~ [0.009] (0.076)	0.020*** [0.005] (0.000)	0.022*** [0.003] (0.000)	0.016*** [0.002] (0.000)	0.015*** [0.002] (0.000)
Lagged post-entry internationalization speed	0.775*** [0.004] (0.000)	0.782*** [0.003] (0.000)	0.867*** [0.001] (0.000)	0.822*** [0.005] (0.000)	0.838*** [0.002] (0.000)	0.850*** [0.001] (0.000)	0.862*** [0.002] (0.000)	0.845*** [0.001] (0.000)
Inverse Mills ratio	-0.001 [0.001] (0.401)	0.000 [0.001] (0.835)	-0.002*** [0.000] (0.000)	-0.003* [0.001] (0.024)	-0.002** [0.001] (0.007)	-0.002*** [0.000] (0.000)	-0.002*** [0.000] (0.000)	-0.002*** [0.000] (0.000)
Constant	0.028 [0.019] (0.137)	0.000 [0.010] (0.995)	0.012** [0.004] (0.002)	-0.004 [0.010] (0.726)	0.013~ [0.007] (0.052)	0.014*** [0.003] (0.000)	0.015*** [0.003] (0.000)	0.003 [0.005] (0.571)
	1.989*** [0.216]	1.645*** [0.099]	1.077*** [0.011]	1.454*** [0.050]	1.207*** [0.023]	1.085*** [0.011]	0.995*** [0.014]	1.146*** [0.009]

(continued on next page)

Table 8 (continued)

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Year dummies	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included
Number of observations	2734.000	2617.000	2617.000	2617.000	2617.000	2617.000	2617.000	2617.000
AR(1)	0.010	0.000	0.000	0.000	0.000	0.000	0.000	0.000
AR(2)	0.061	0.103	0.215	0.073	0.173	0.099	0.205	0.119
Hansen	16.549	86.768	207.075	71.794	140.423	196.476	178.301	183.712
P-value of Hansen J statistics	0.485	0.784	1.000	0.758	0.569	0.998	1.000	0.995

Notes: (1) $\sim p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed tests); (2) standard errors in parentheses; p-values in square brackets

Table 9

Summary of selection and nescience effects.

Effect Type	Variable	Measure	Low Big Data Analytics	High Big Data Analytics	Change	Interpretation
Selection effect	Depth	Curvature (Depth squared $\times$ Big Data Analytics coefficient)	-0.0053	0.003	0.0083	Inverted U $\rightarrow$ U-shaped
Selection effect	Breadth	Curvature (Breadth squared $\times$ Big Data Analytics coefficient)	0.5495	0.2543	-0.2952	U-shape flattens 54%
Nescience effect	Depth	Turning point	-2.4814	-0.27	2.2114	89% shift in optimal
Nescience effect	Breadth	Turning point	0.2233	0.812	0.5887	264% increase in optimal

decisions benefit from algorithmic optimization, and which require contextual interpretation. Big Data Analytics is most effective for decisions grounded in regularity and scale, such as global demand forecasting, risk scoring, or optimizing logistics networks, where its statistical advantages outweigh the need for contextual nuance. Decisions involving institutional ambiguity, informal norms, and relational embeddedness, such as partner negotiations, regulatory interpretation, and cultural adaptation, should remain anchored in managerial expertise. Clear governance norms help firms avoid the twin hazards of algorithmic overreach and experiential bias.

Our findings also carry broader policy implications, particularly in relation to the digital infrastructures that support responsible internationalization. Policymakers play an essential role in ensuring that firms have access to high-quality, interoperable data while maintaining strong safeguards around privacy, transparency, and algorithmic accountability. Policies that promote fair data access, cross-border data governance, and responsible AI standards enable firms to internationalize more effectively without exacerbating information asymmetries or institutional risks. Additionally, public-private partnerships focused on digital skills development can ensure that managerial workforces possess the hybrid competencies needed to integrate Big Data Analytics with experiential judgment - an increasingly central capability in global strategy.

Finally, our insights also align with and support several United Nations Sustainable Development Goals (SDGs). By showing how firms can use Big Data Analytics responsibly to enhance adaptive learning, our study contributes to SDG 8 (Decent Work and Economic Growth) by informing more resilient, efficient, and sustainable international expansion strategies that promote stable employment and productivity growth in host markets. Moreover, the emphasis on balancing data-driven decision-making with contextual sensitivity resonates with SDG 9 (Industry, Innovation, and Infrastructure), as it illustrates how digital technologies can complement human experiential learning to foster more innovative, evidence-based, and institutionally aware global operations.

6.3. Limitations

This study has limitations that open avenues for future research. First, its reliance on secondary data from U.S-listed firms may constrain generalizability. Expanding to cross-national samples, especially in

emerging markets, could test whether the interplay between Big Data Analytics and experiential learning holds across contexts. Second, we exclude IT-industry firms because their inherently high digital intensity makes Big Data Analytics a built-in component of core operations rather than a distinct, observable capability, thus limiting the generalizability of our findings to more traditional, non-IT industries. Future research could explicitly compare IT and non-IT sectors to assess whether the learning dynamics uncovered here operate differently when Big Data Analytics constitutes an industry norm. Third, Big Data Analytics implementation is proxied by public announcements of strategic initiatives, which may not capture the quality, sophistication, or integration of Big Data Analytics capabilities. More granular, survey-based measures could address this gap. Finally, our quantitative design cannot directly observe the cognitive and decision-making processes through which managers reconcile data-driven insights with experiential knowledge. Future research could employ qualitative case studies to provide richer, process-oriented insights into how managers reconcile human intuition with Big Data Analytics-driven learning.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ibusrev.2026.102587](https://doi.org/10.1016/j.ibusrev.2026.102587).

Data Availability

Data will be made available on request.

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